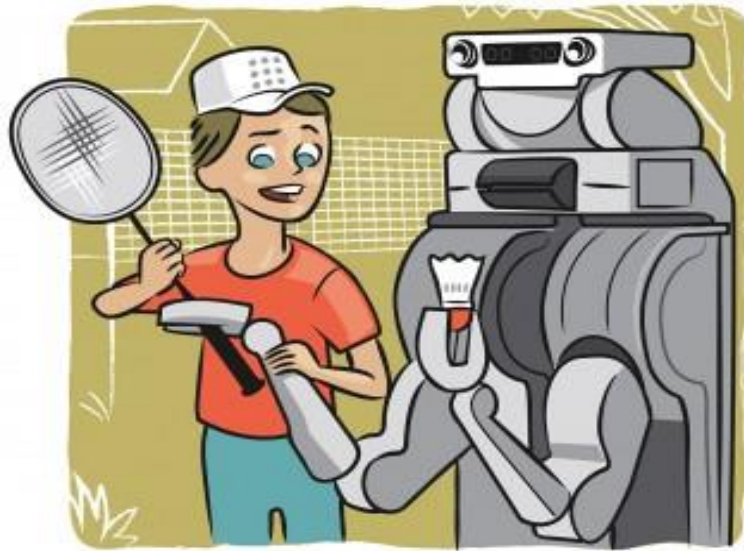


Imitation Learning and Inverse RL

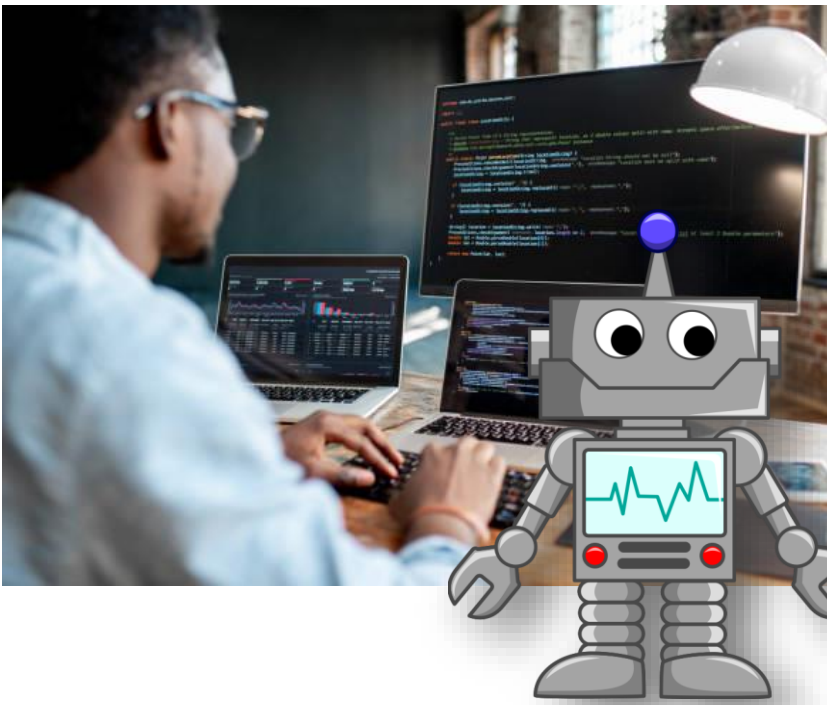


Instructor: Daniel Brown

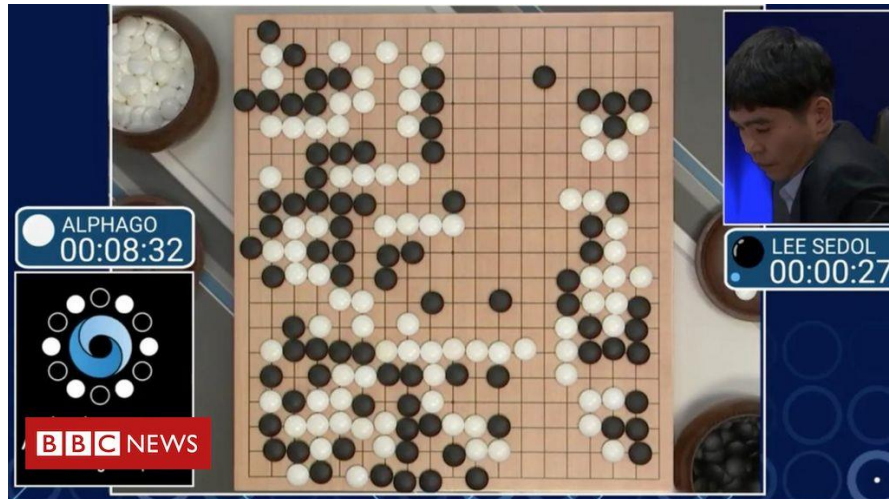
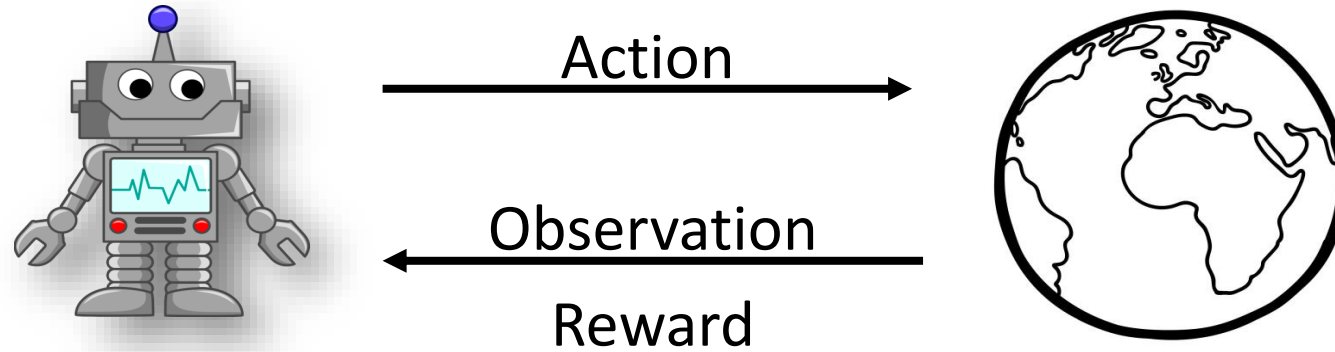
[Some slides adapted from Sergey Levine (CS 285) and Alina Vereshchaka (CSE4/510), and Ryan Hoque]

Announcements

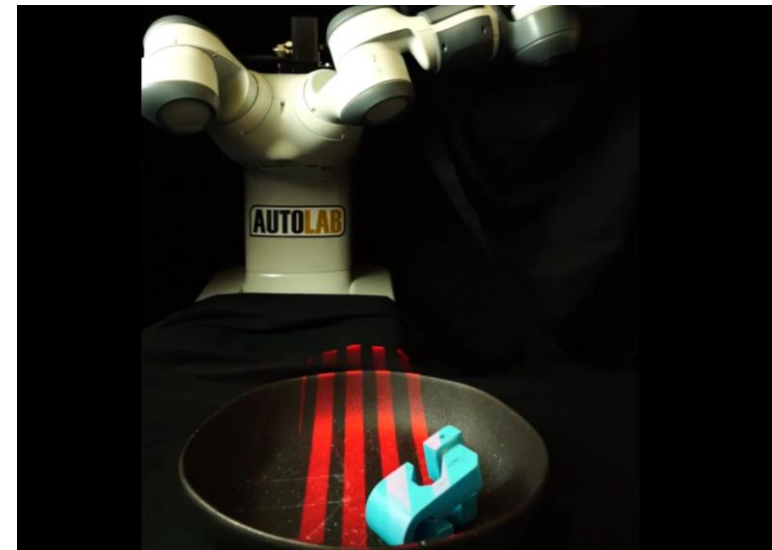
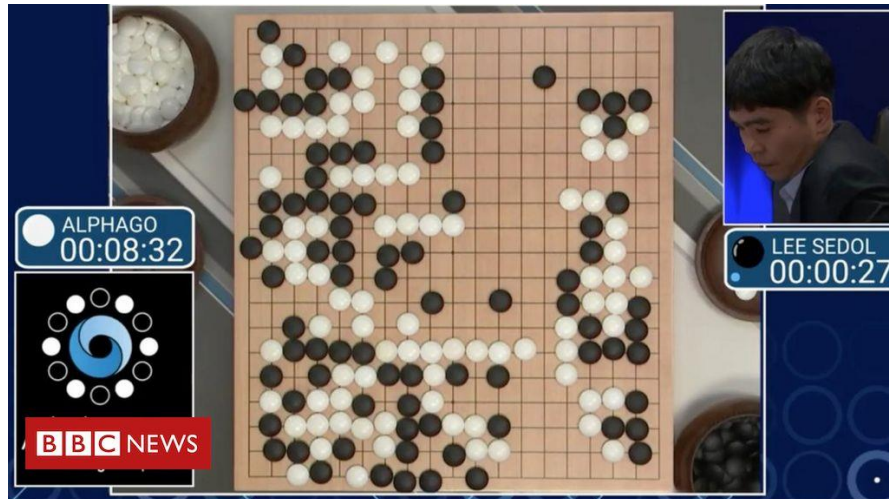
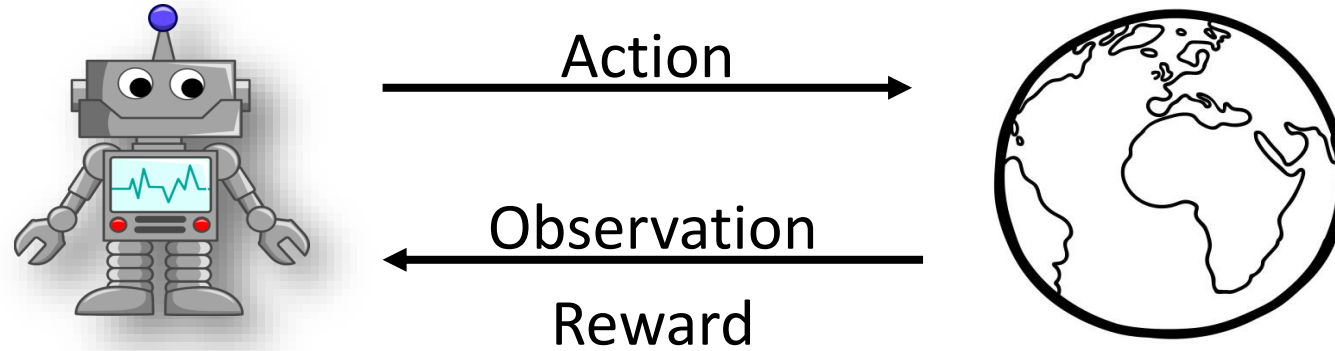
- HW 8 due tomorrow!
- Get started on Project 5.
- Check grades on gradescope and canvas to make sure they are consistent and show up on canvas. If not, visit a TA during office hours or make a post on Piazza.



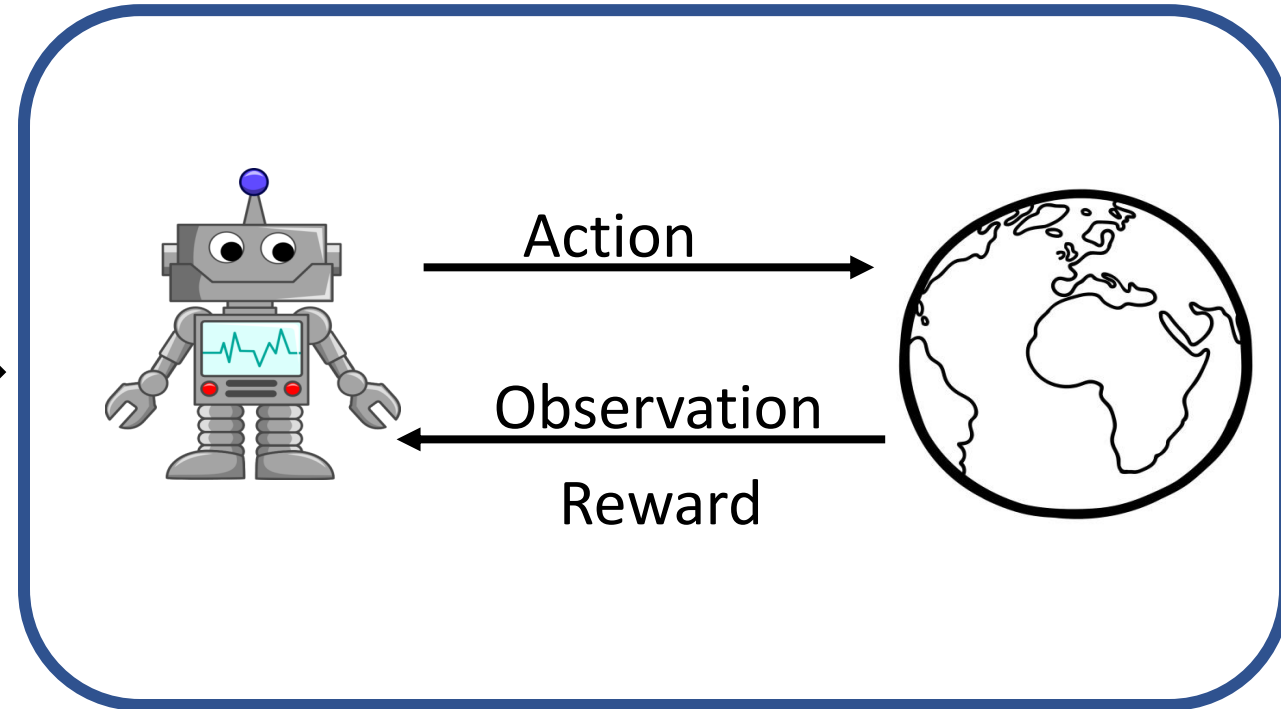
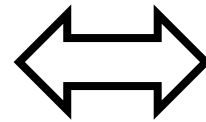
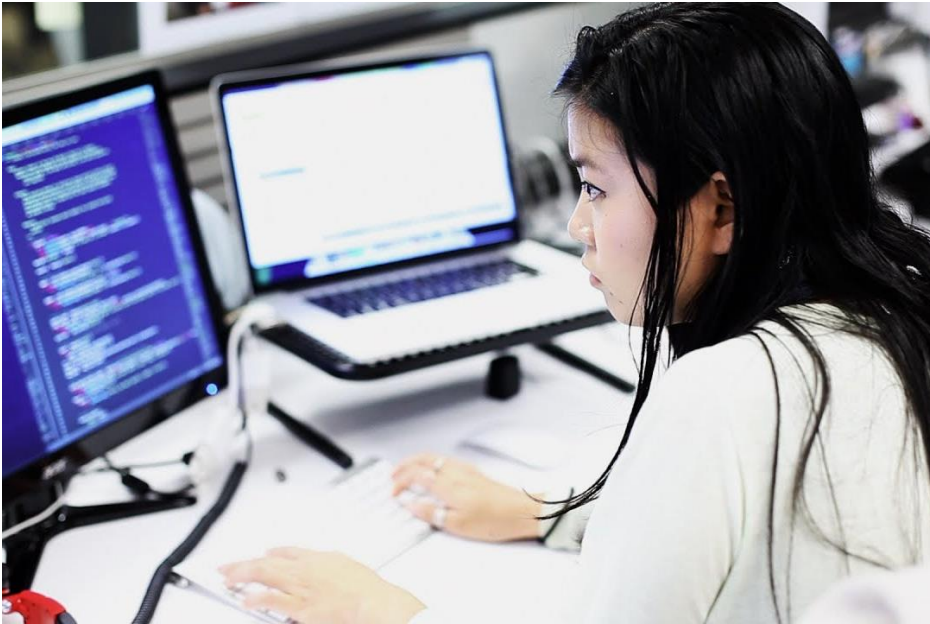
Reinforcement Learning



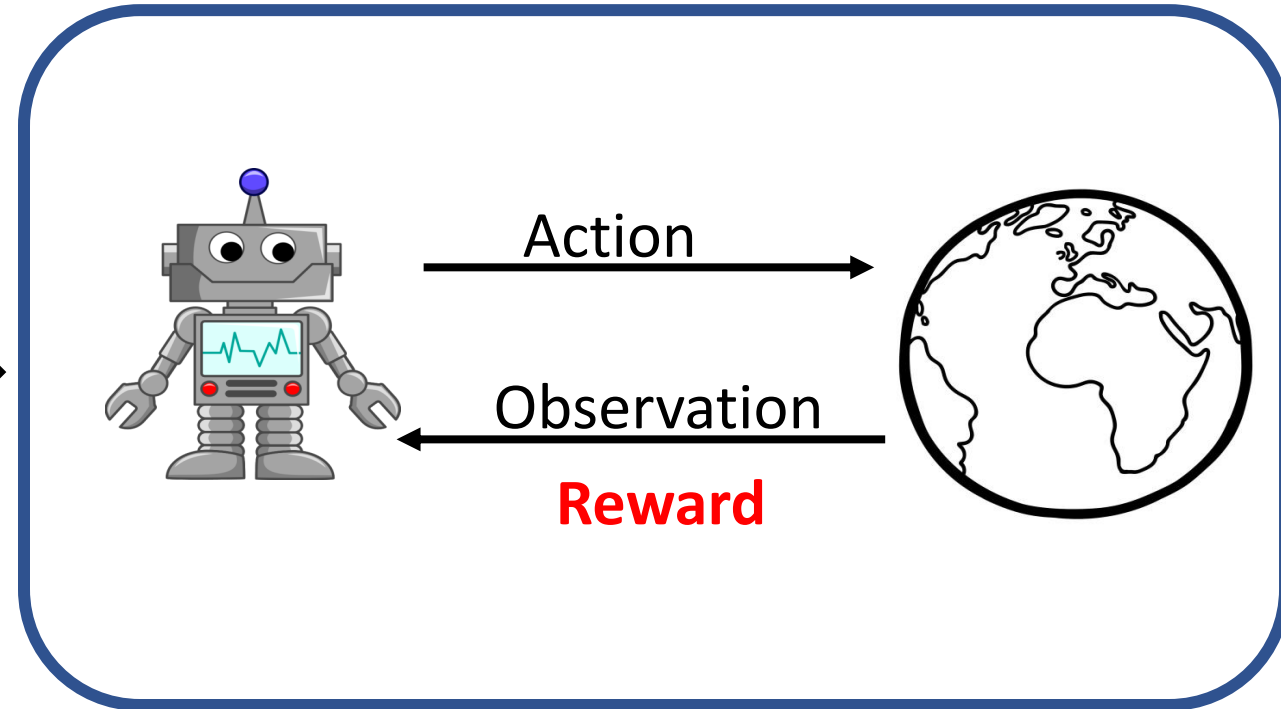
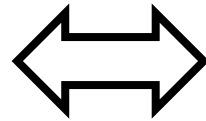
Reinforcement Learning



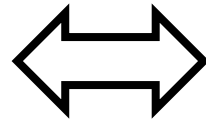
Reward engineering is hard!



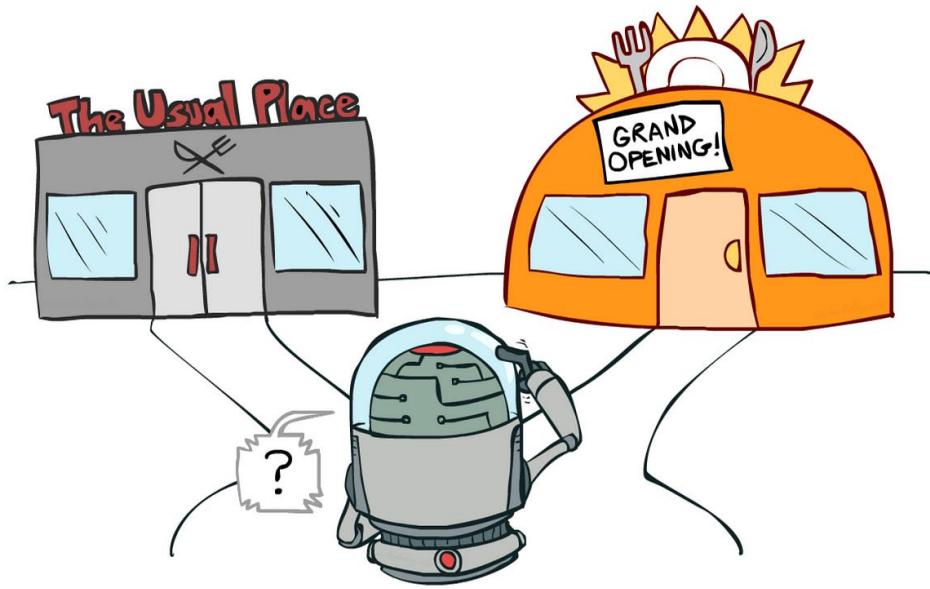
Reward engineering is hard!



Reward engineering is hard!



Reinforcement learning is hard...even with a reward function!



Imitation Learning:

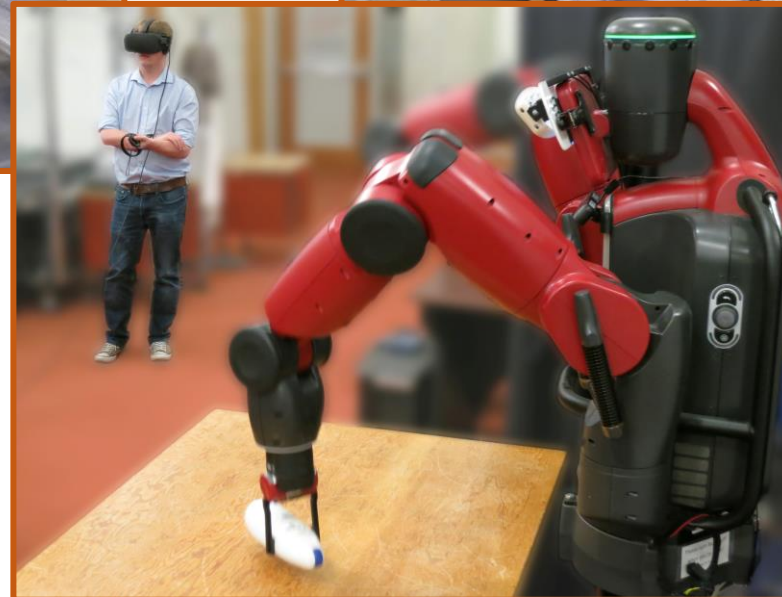
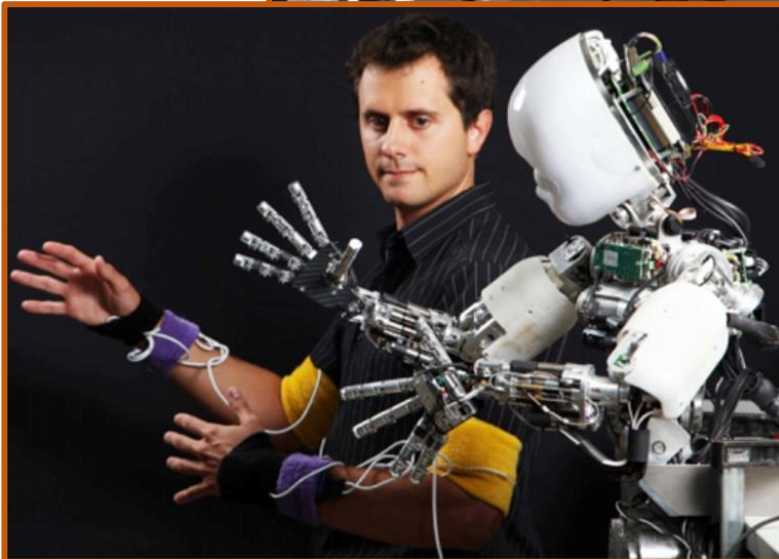
Learn a policy from examples of good behavior.



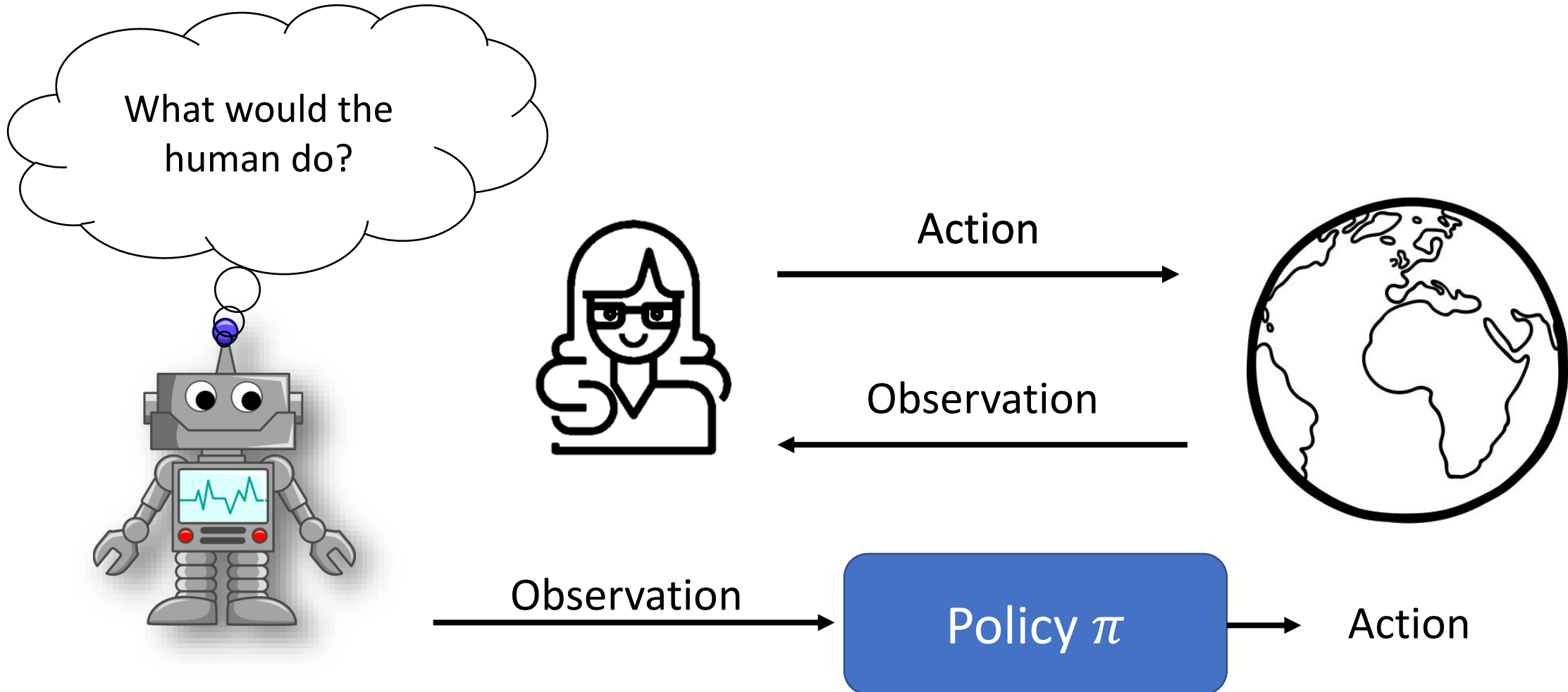
- Often showing is easier than telling.
- Alleviates problem of exploration.



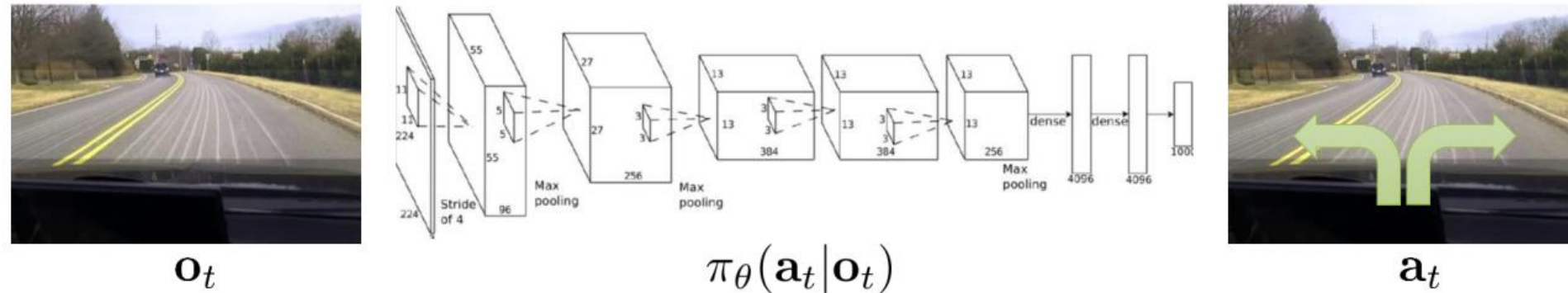
Learning From Demonstration (LfD)



Behavioral Cloning

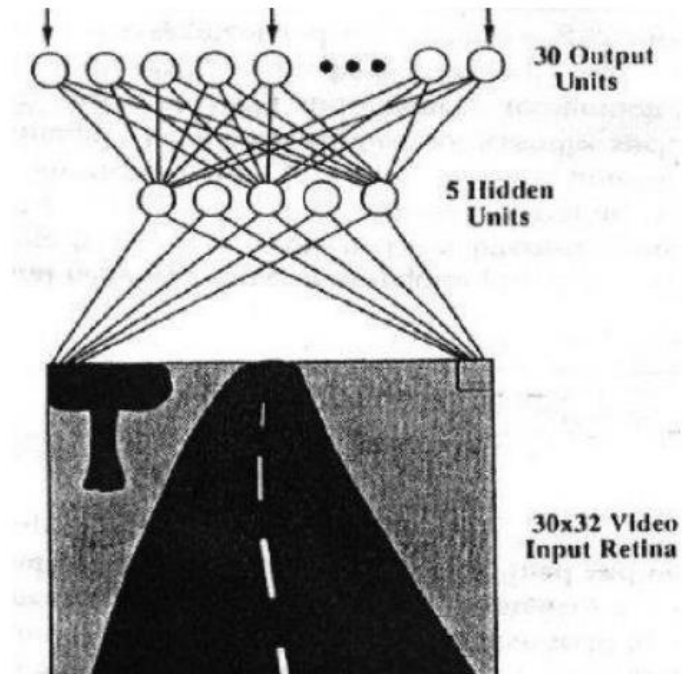


Imitation Learning via Behavioral Cloning



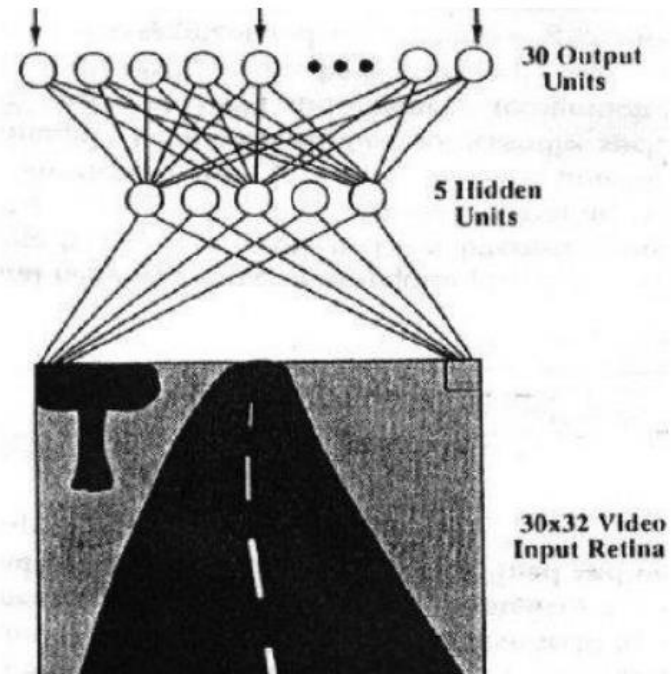
ALVINN: One of the first imitation learning systems

ALVINN: **A**utonomous **L**and **V**ehicle **I**n a **N**eural **N**etwork
1989

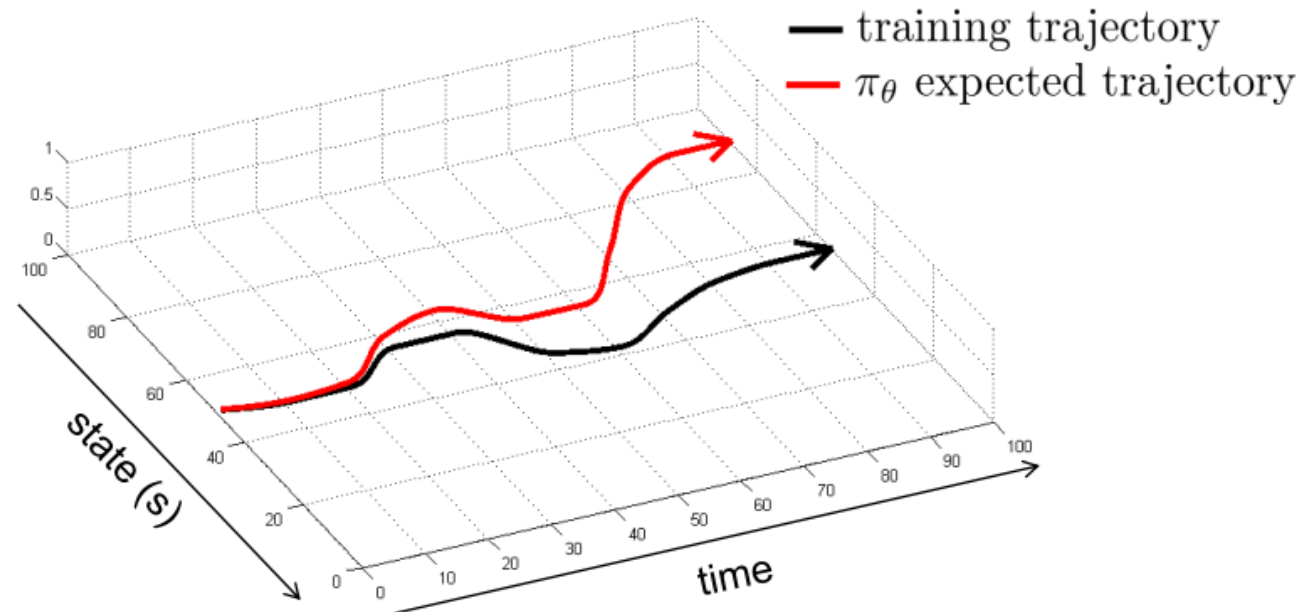
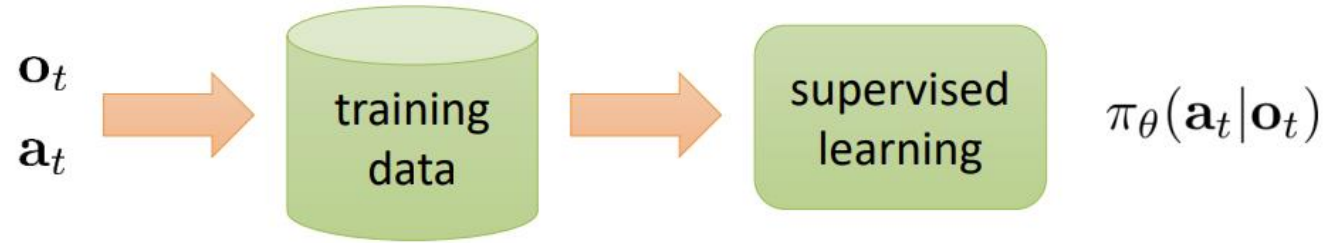


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1989

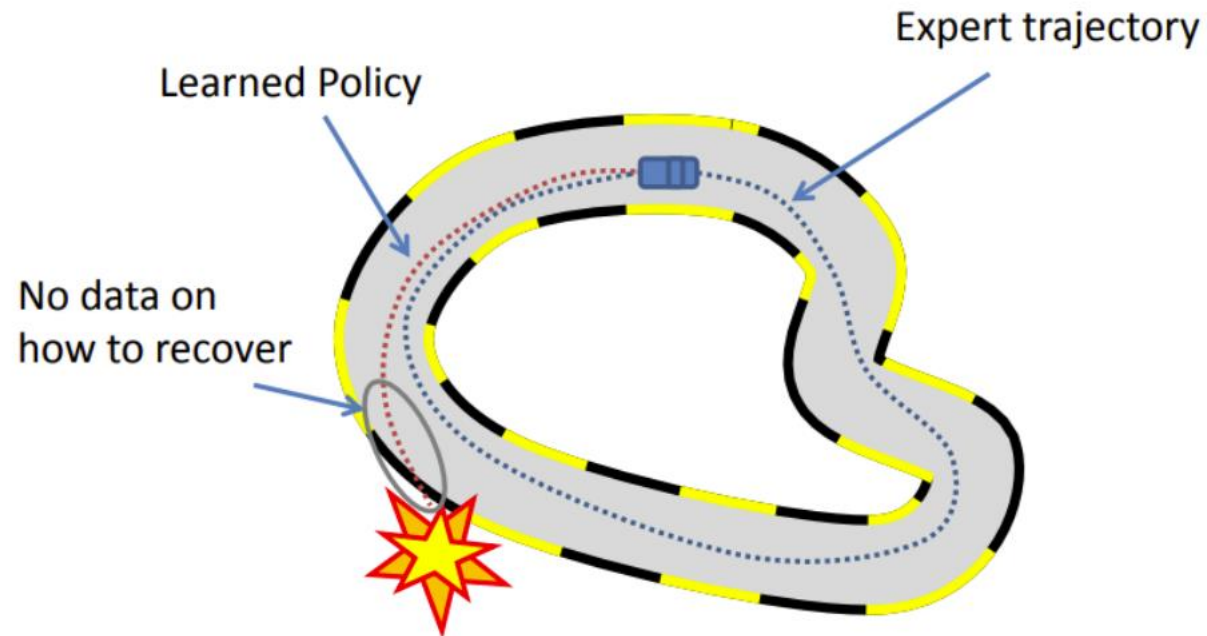


What could go wrong?



Distribution Shift

$$p_{\pi^*}(o_t) \neq p_{\pi_\theta}(o_t)$$

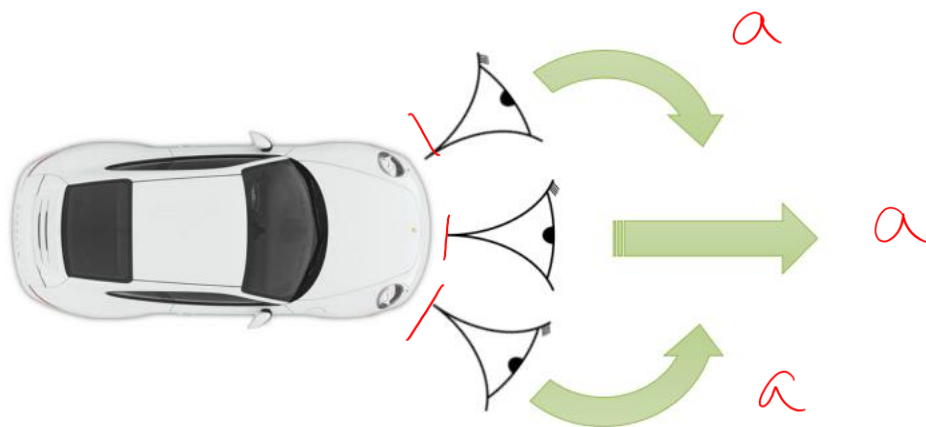
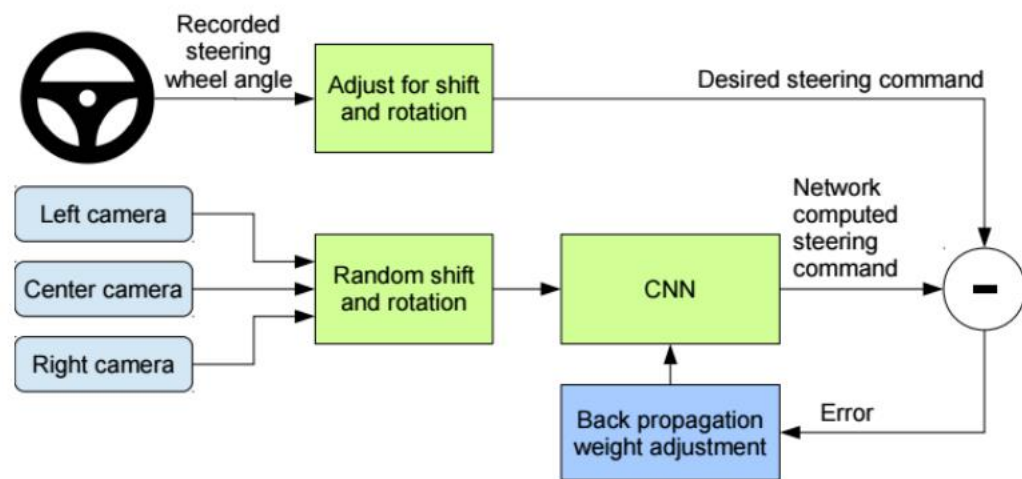


	Supervised Learning	Supervised Learning + Control
Train	$(x, y) \sim D$	$s \sim P(\cdot s, \pi^*(s))$
Test	$(x, y) \sim D$	$s \sim P(\cdot s, \pi_\theta(s))$

But it still can work in practice...

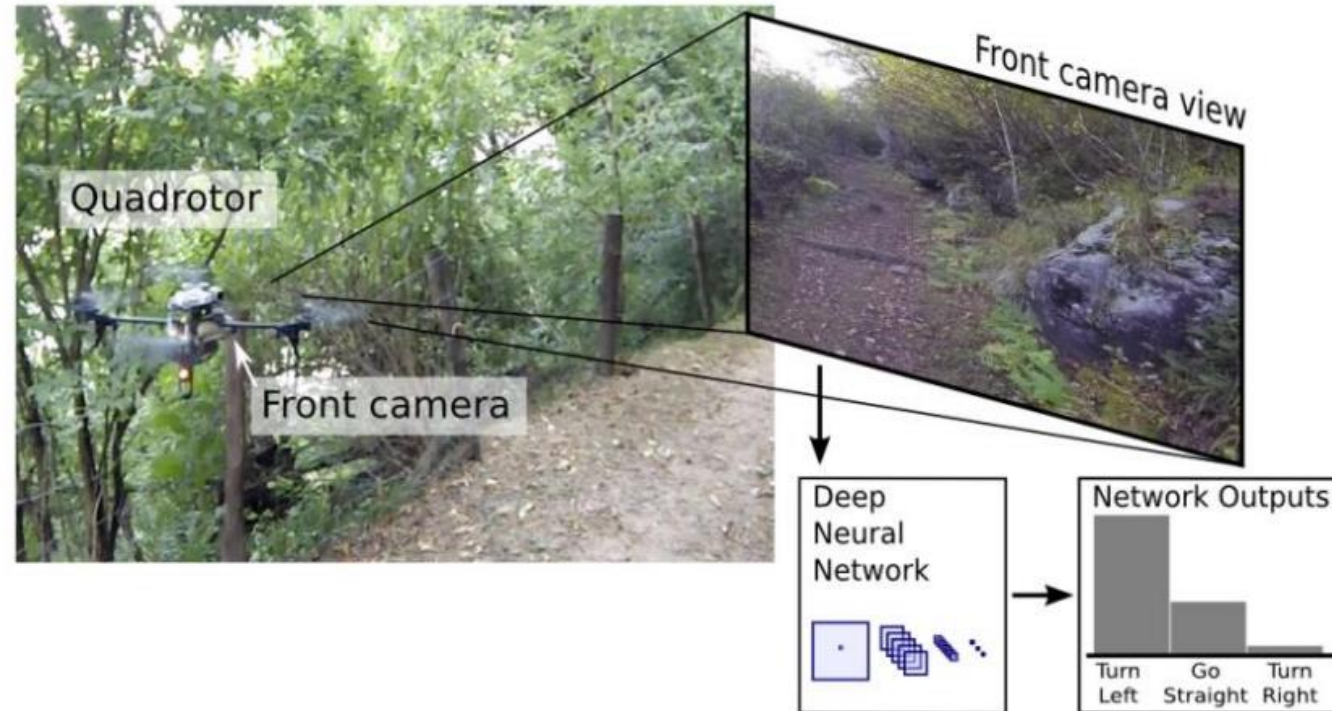


How?



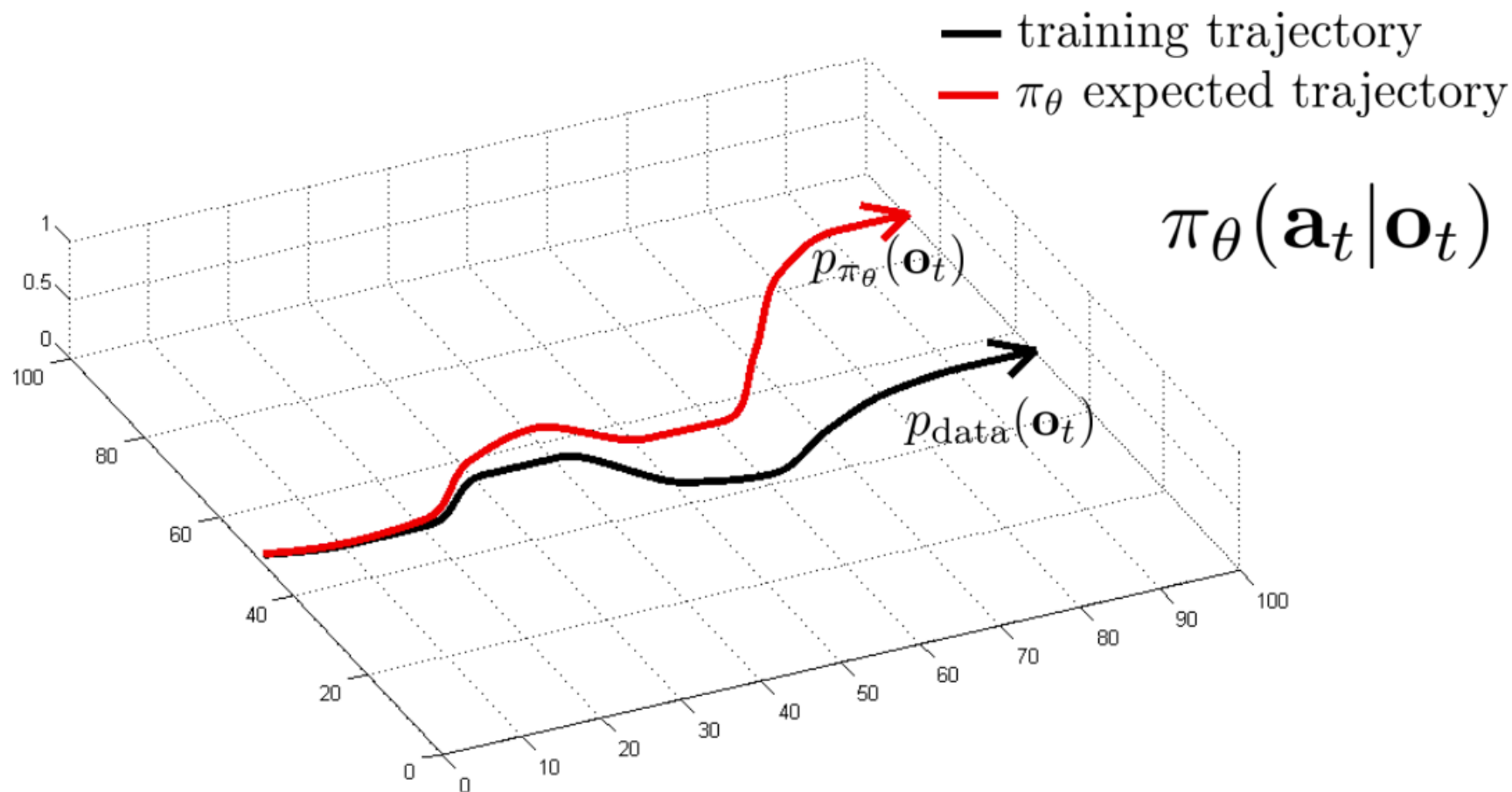
A Machine Learning Approach to Visual Perception of Forest Trails for Mobile Robots

Alessandro Giusti¹, Jérôme Guzzi¹, Dan C. Cireşan¹, Fang-Lin He¹, Juan P. Rodríguez¹
Flavio Fontana², Matthias Faessler², Christian Forster²
Jürgen Schmidhuber¹, Gianni Di Caro¹, Davide Scaramuzza², Luca M. Gambardella¹



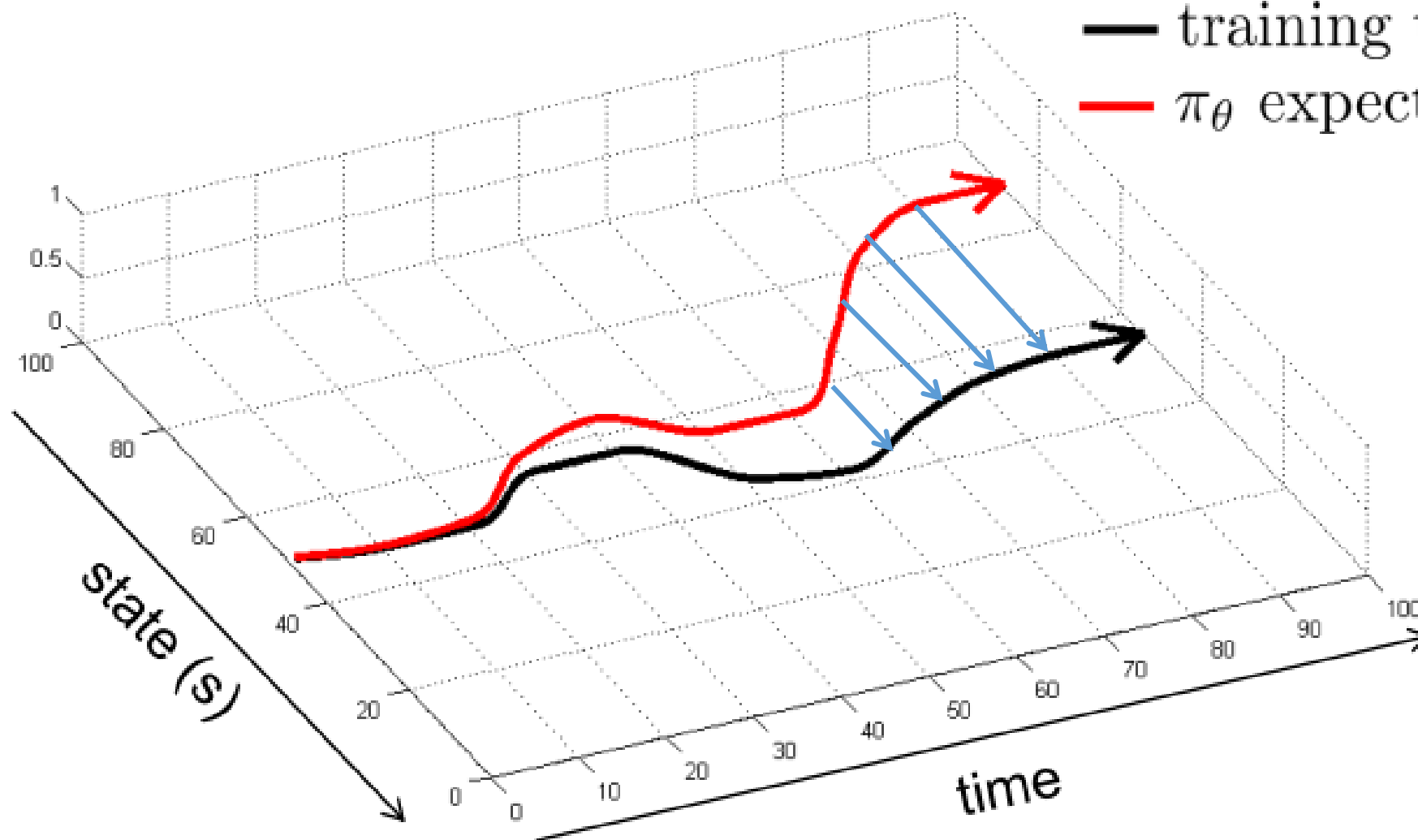


Can we make it work more often?



can we make $p_{\text{data}}(\mathbf{o}_t) = p_{\pi_\theta}(\mathbf{o}_t)$?

- human recovery policy
- training trajectory
- π_θ expected trajectory



DAgger

can we make $p_{\text{data}}(\mathbf{o}_t) = p_{\pi_\theta}(\mathbf{o}_t)$?

idea: instead of being clever about $p_{\pi_\theta}(\mathbf{o}_t)$, be clever about $p_{\text{data}}(\mathbf{o}_t)$!

DAgger: Dataset Aggregation

goal: collect training data from $p_{\pi_\theta}(\mathbf{o}_t)$ instead of $p_{\text{data}}(\mathbf{o}_t)$

how? just run $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$

but need labels $\mathbf{a}_t$!

- 
1. train $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ from human data $\mathcal{D} = \{\mathbf{o}_1, \mathbf{a}_1, \dots, \mathbf{o}_N, \mathbf{a}_N\}$
 2. run $\pi_\theta(\mathbf{a}_t|\mathbf{o}_t)$ to get dataset $\mathcal{D}_\pi = \{\mathbf{o}_1, \dots, \mathbf{o}_M\}$
 3. Ask human to label $\mathcal{D}_\pi$ with actions $\mathbf{a}_t$
 4. Aggregate: $\mathcal{D} \leftarrow \mathcal{D} \cup \mathcal{D}_\pi$

DAgger has very nice theoretical guarantees.

Why might it be **hard** to implement in practice?

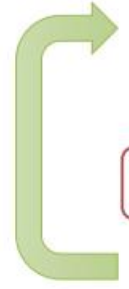
DAgger: **D**ataset **A**ggregation

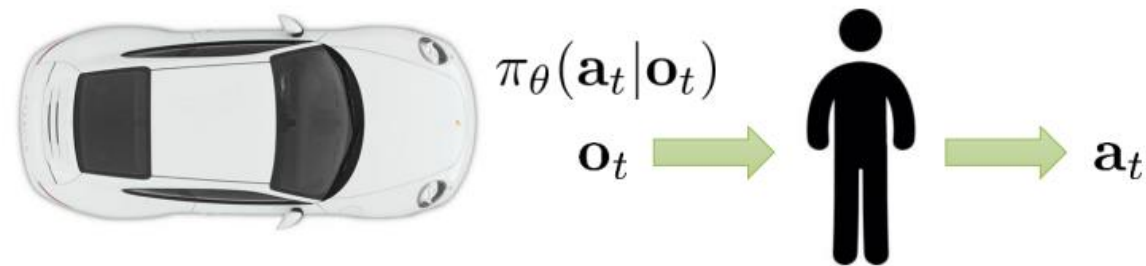
goal: collect training data from $p_{\pi_{\theta}}(\mathbf{o}_t)$ instead of $p_{\text{data}}(\mathbf{o}_t)$

how? just run $\pi_{\theta}(\mathbf{a}_t|\mathbf{o}_t)$

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- 
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DAgger has very nice theoretical guarantees.

Why might it be **easy** to implement in practice?

DAgger: **D**ataset **A**ggregation

goal: collect training data from $p_{\pi_{\theta}}(\mathbf{o}_t)$ instead of $p_{\text{data}}(\mathbf{o}_t)$

how? just run $\pi_{\theta}(\mathbf{a}_t|\mathbf{o}_t)$

but need labels $\mathbf{a}_t$!

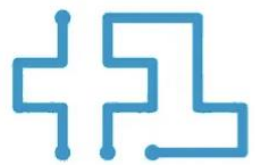
- 
1. train $\pi_{\theta}(\mathbf{a}_t|\mathbf{o}_t)$ from human data $\mathcal{D} = \{\mathbf{o}_1, \mathbf{a}_1, \dots, \mathbf{o}_N, \mathbf{a}_N\}$
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 3. Ask human to label $\mathcal{D}_{\pi}$ with actions $\mathbf{a}_t$
 4. Aggregate: $\mathcal{D} \leftarrow \mathcal{D} \cup \mathcal{D}_{\pi}$

Learn from an Algorithmic Supervisor!



But we don't always have access to an algorithmic supervisor...

Can we make DAgger more practical when dealing with real human labeling?



PLUS ONE
ROBOTICS



WAYMO

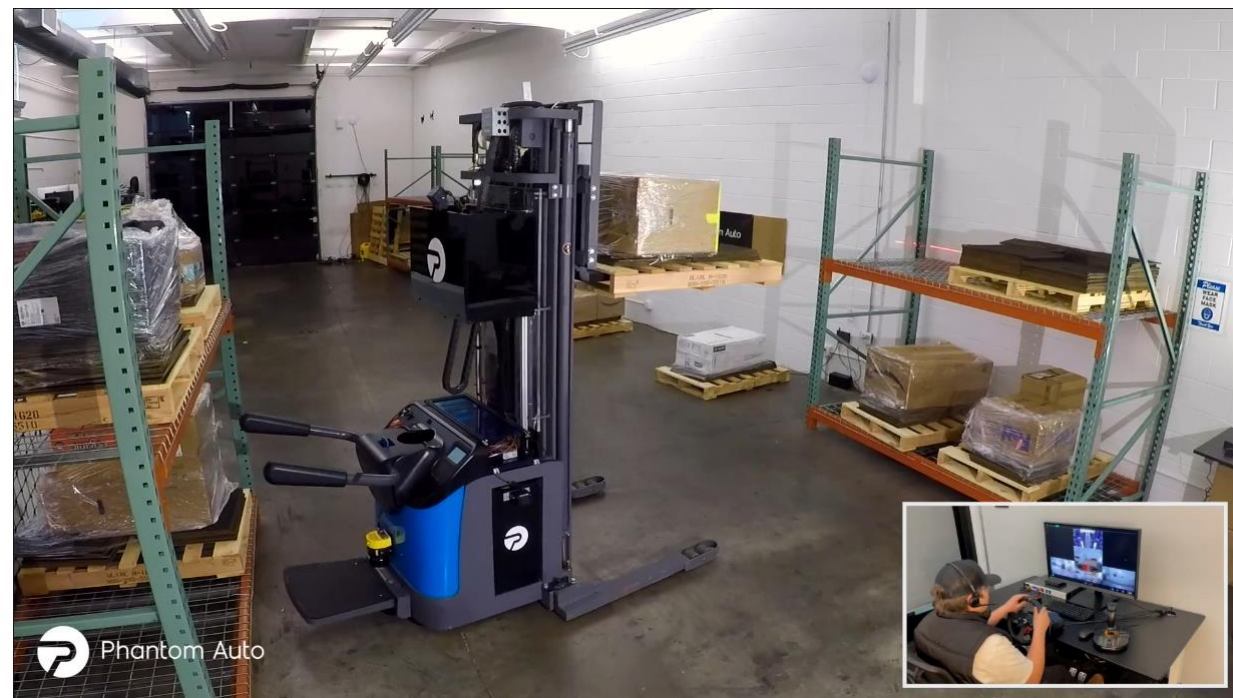
ZOOX



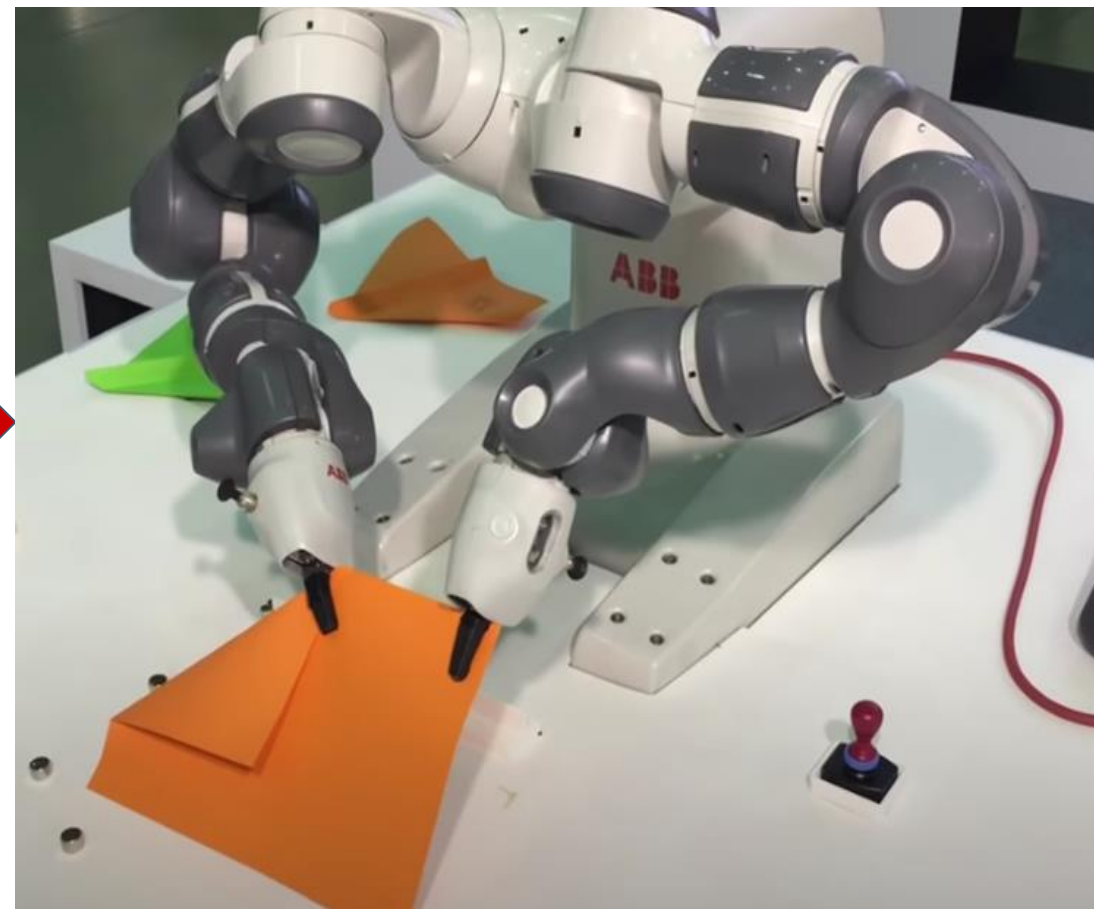
nimble



Phantom Auto



Interactive IL



Interactive IL

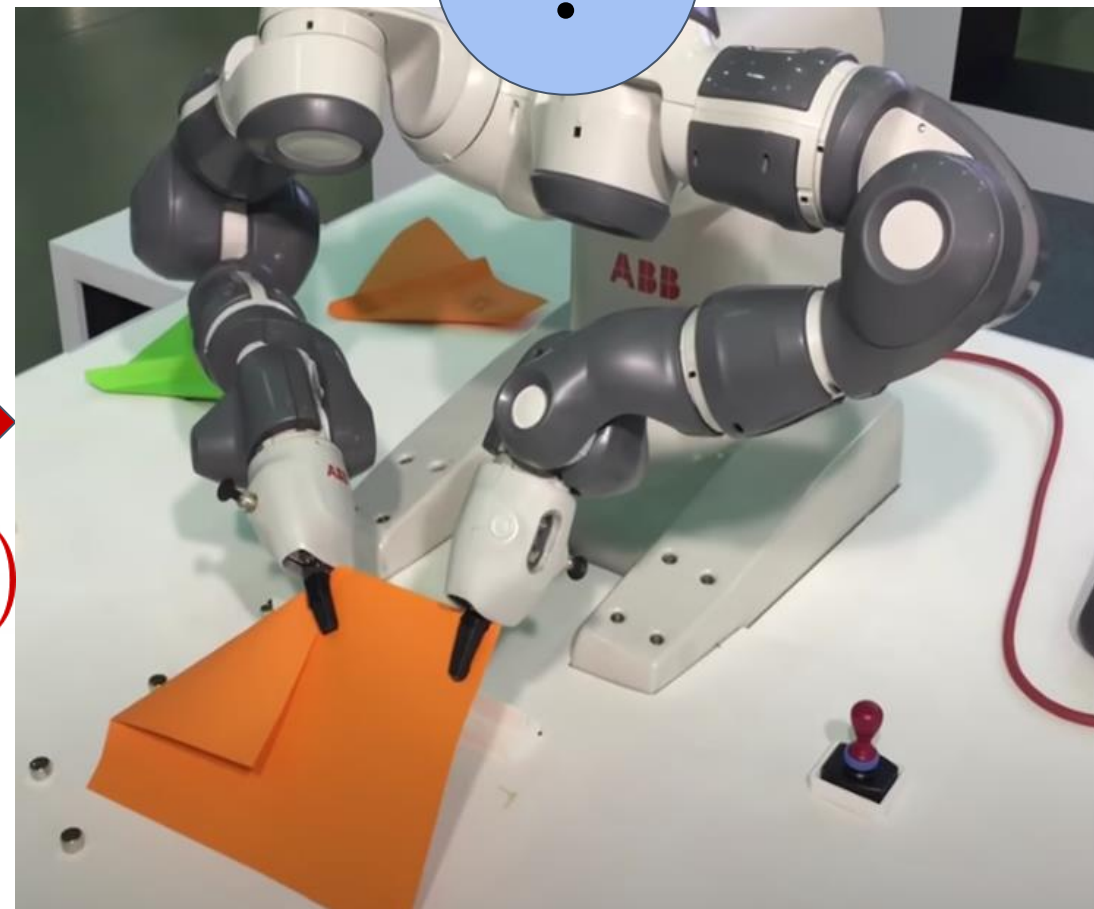


$$\pi_H(s)$$



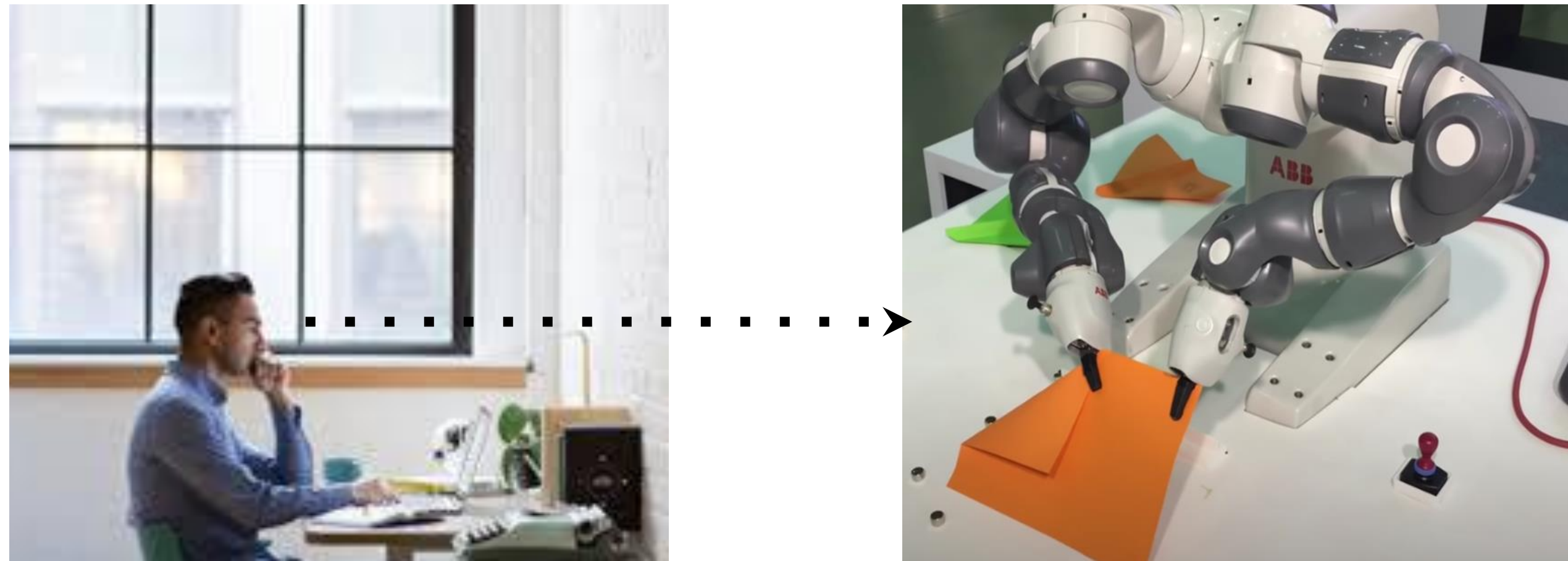
$$\pi_{\text{meta}}(s)$$

???



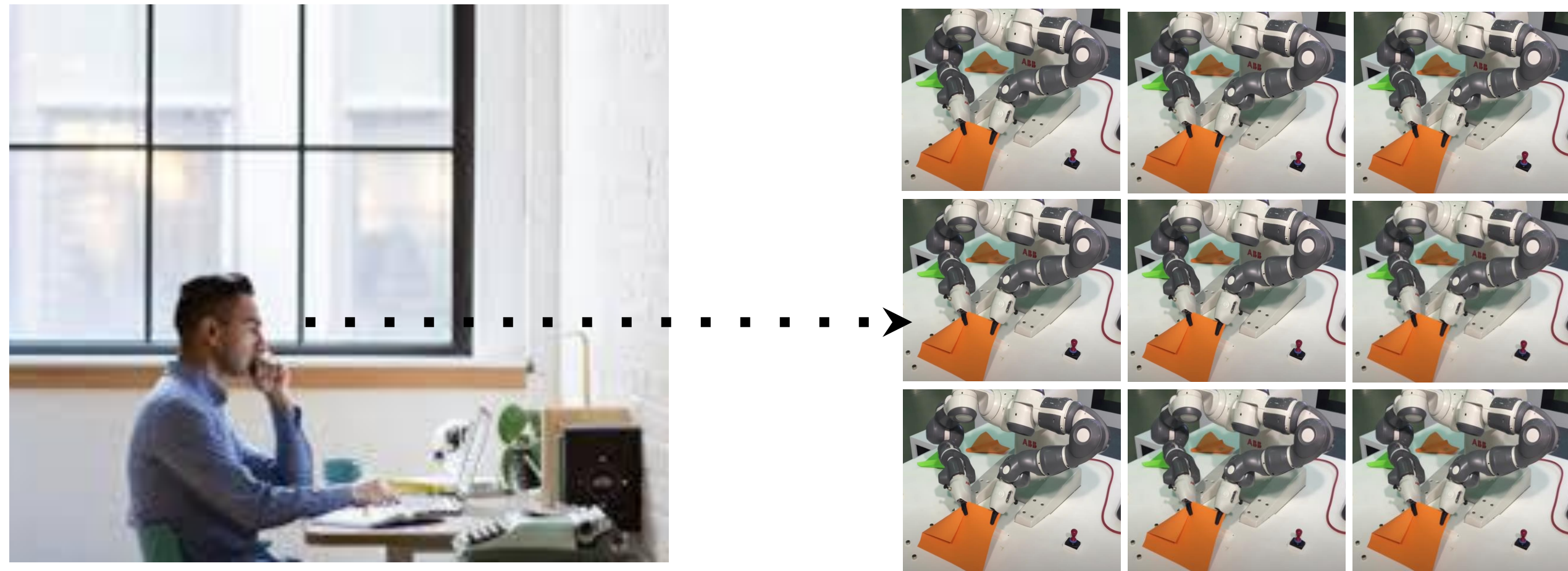
$$\pi_R(s)$$

Human-Gated Interactive IL



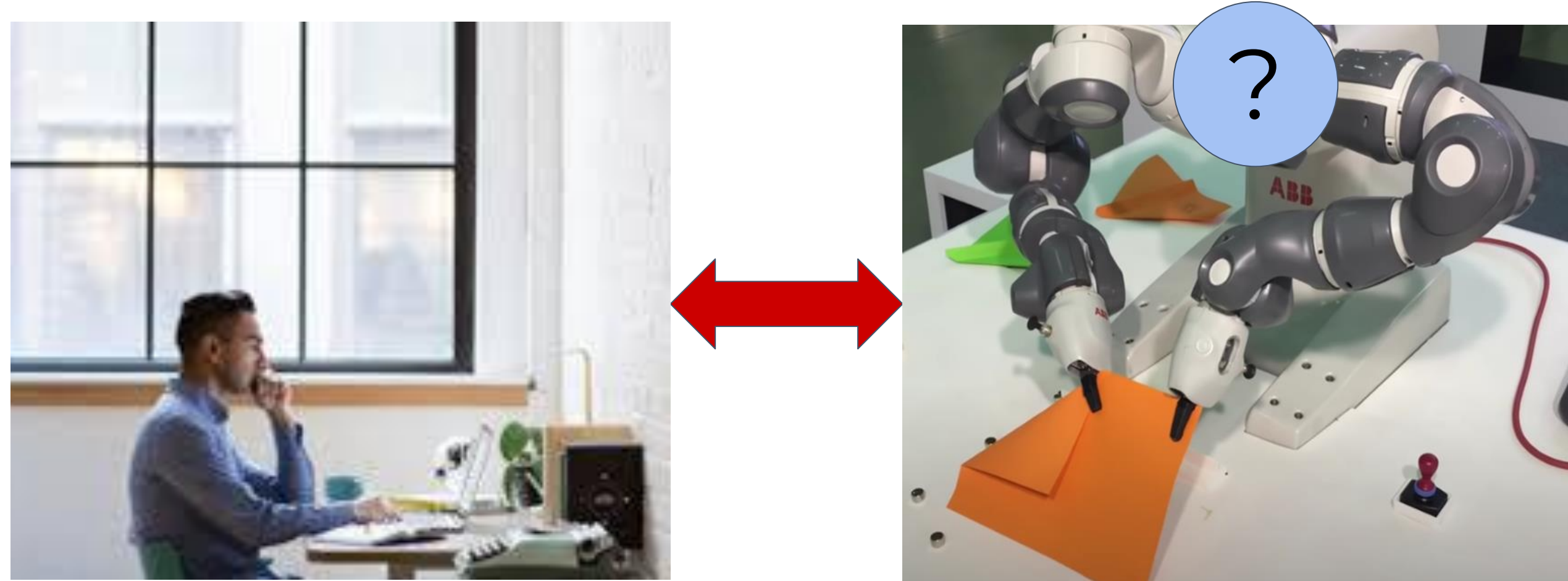
[3] M. Kelly, C. Sidrane, K. Driggs-Campbell, and M. J. Kochenderfer. HG-Dagger: Interactive Imitation Learning with Human Experts. ICRA 2019.

Human-Gated Interactive IL



[3] M. Kelly, C. Sidrane, K. Driggs-Campbell, and M. J. Kochenderfer. HG-Dagger: Interactive Imitation Learning with Human Experts. ICRA 2019.

Robot-Gated Interactive IL



- [4] J. Zhang, K. Cho. Query-Efficient Imitation Learning for End-to-End Autonomous Driving. AAAI 2017.
- [5] K. Menda, K. Driggs-Campbell, M. Kochenderfer. EnsembleDAgger: A Bayesian Approach to Safe Imitation Learning. IROS 2019.

When should a robot ask for help?



Novel (and risky)

When should a robot ask for help?

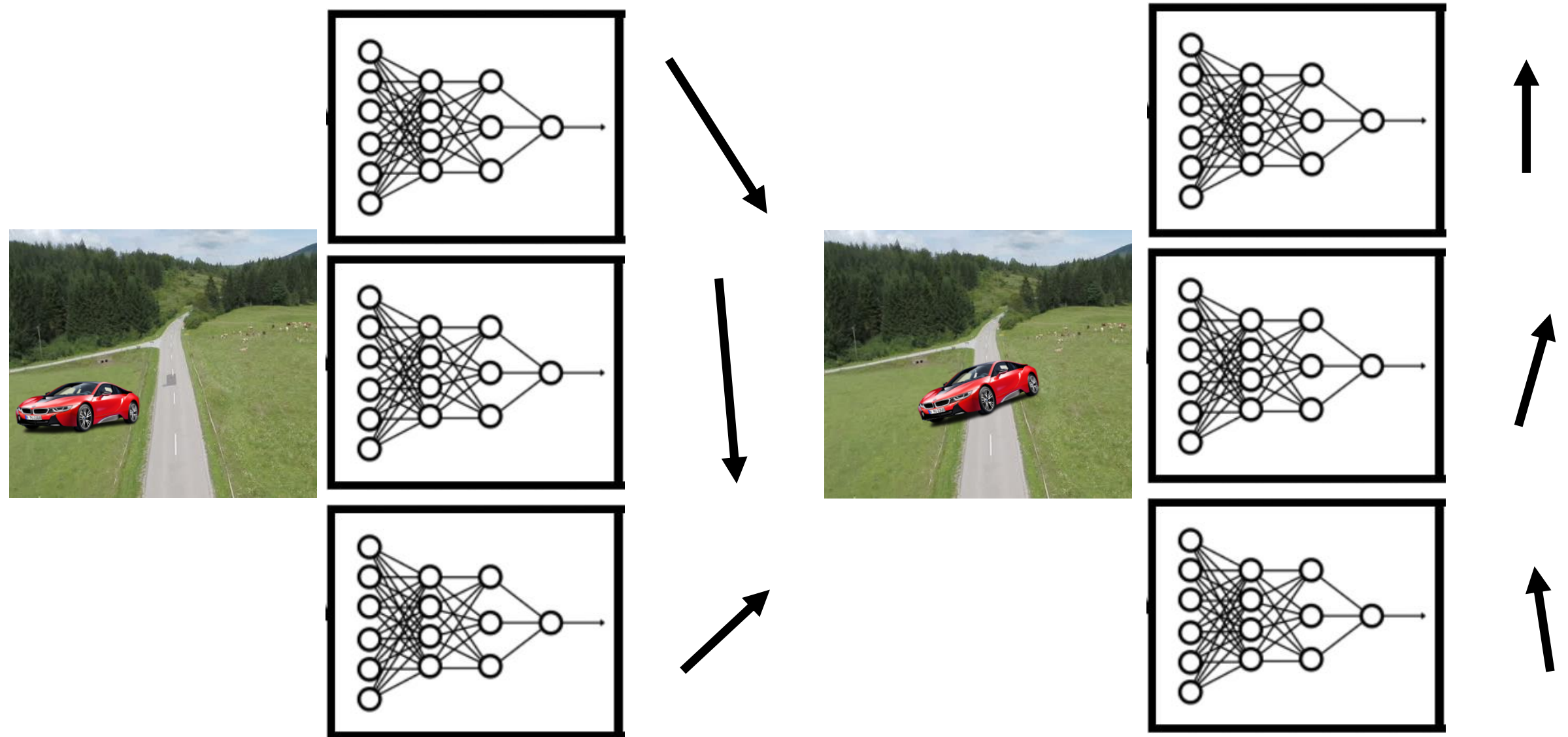


Novel (and risky)

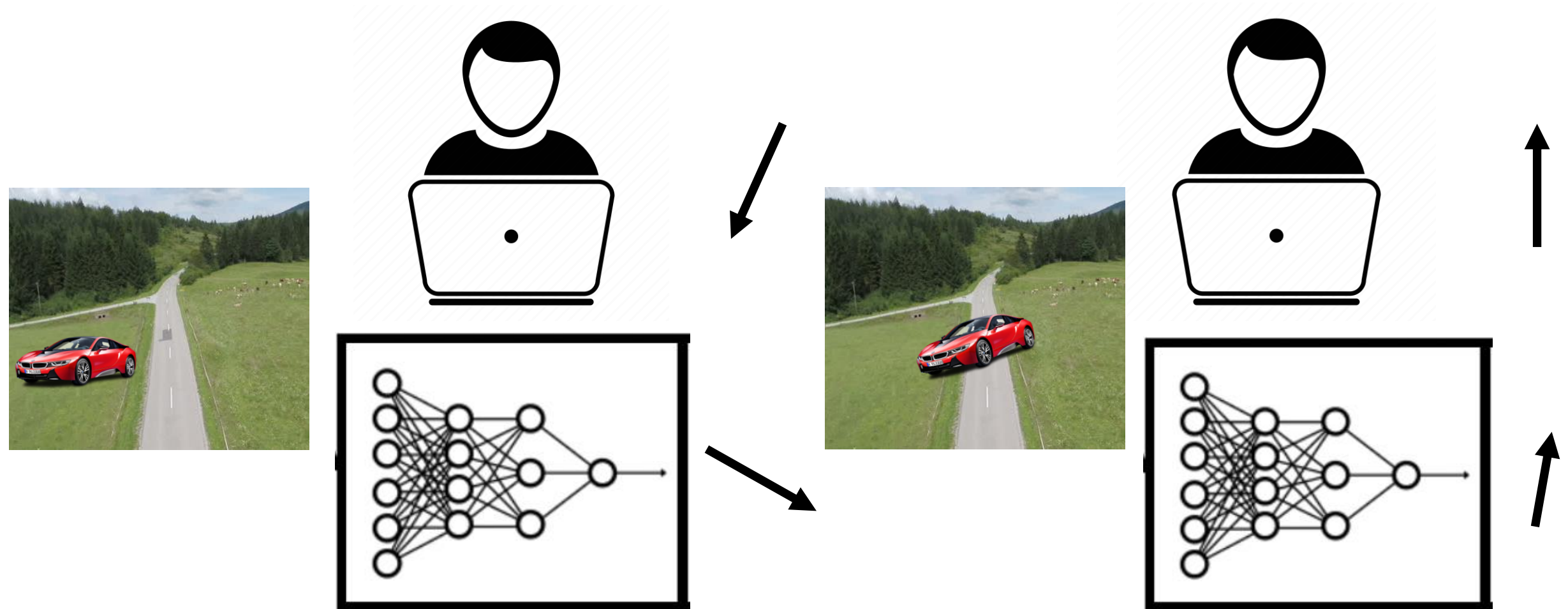


Risky (but not novel)

Novelty Estimation



Novelty Estimation: Supervisor Mode



Risk Estimation

$$Q_{\mathcal{G}}^{\pi_r}(s_t, a_t) = \mathbb{E}_{\pi_r} \left[\sum_{t'=t}^{\infty} \gamma^{t'-t} \mathbb{1}_{\mathcal{G}}(s'_t) | s_t, a_t \right]$$

Risk Estimation

$$Q_{\mathcal{G}}^{\pi_r}(s_t, a_t) = \mathbb{E}_{\pi_r} \left[\sum_{t'=t}^{\infty} \gamma^{t'-t} \mathbb{1}_{\mathcal{G}}(s'_t) | s_t, a_t \right]$$

$$\text{Risk}^{\pi_r}(s, a) = 1 - \hat{Q}_{\phi, \mathcal{G}}^{\pi_r}(s, a)$$

Risk Estimation

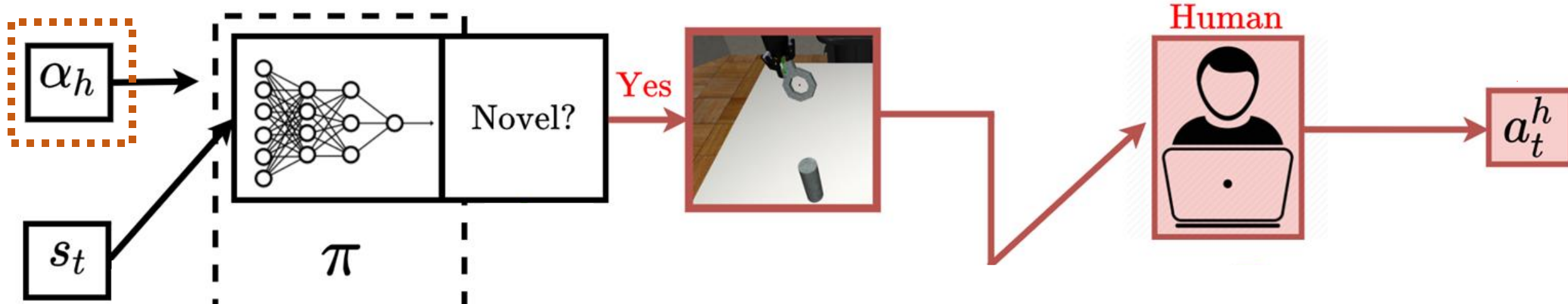
$$Q_{\mathcal{G}}^{\pi_r}(s_t, a_t) = \mathbb{E}_{\pi_r} \left[\sum_{t'=t}^{\infty} \gamma^{t'-t} \mathbb{1}_{\mathcal{G}}(s'_t) | s_t, a_t \right]$$

$$\text{Risk}^{\pi_r}(s, a) = 1 - \hat{Q}_{\phi, \mathcal{G}}^{\pi_r}(s, a)$$

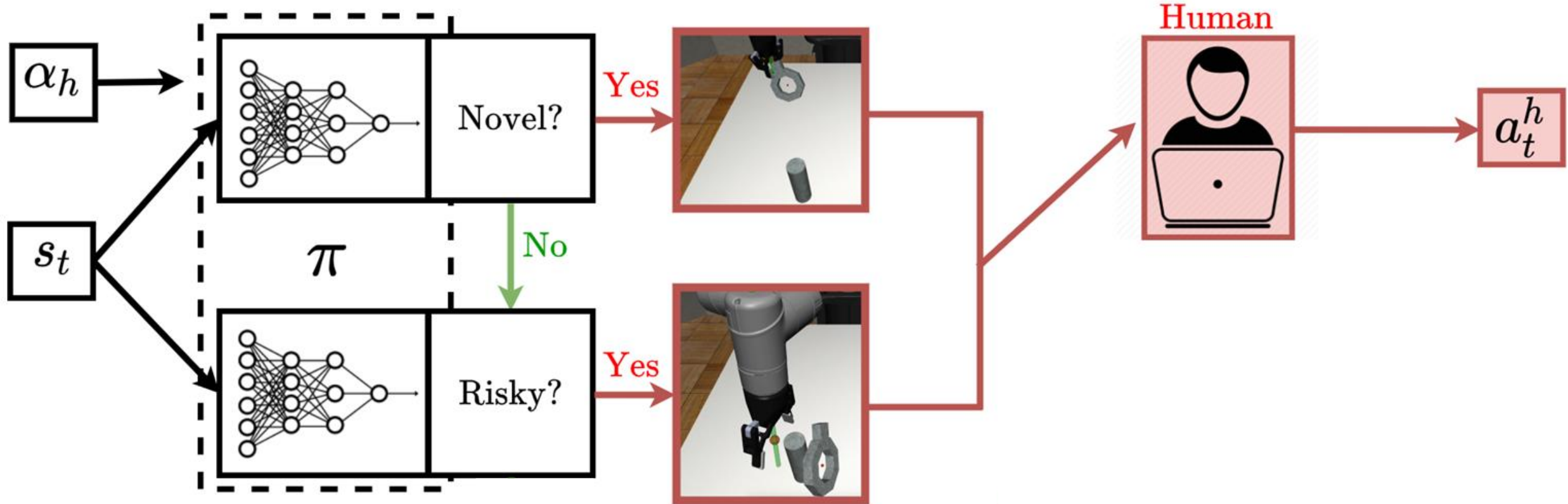
$$J_{\mathcal{G}}^Q(s_t, a_t, s_{t+1}; \phi) = \frac{1}{2} \left(\hat{Q}_{\phi, \mathcal{G}}^{\pi_r}(s_t, a_t) - (\mathbb{1}_{\mathcal{G}}(s_t) + (1 - \mathbb{1}_{\mathcal{G}}(s_t))\gamma \hat{Q}_{\phi, \mathcal{G}}^{\pi_r}(s_{t+1}, \pi_r(s_{t+1}))) \right)^2$$

Target percent of time human
wants to give interventions.

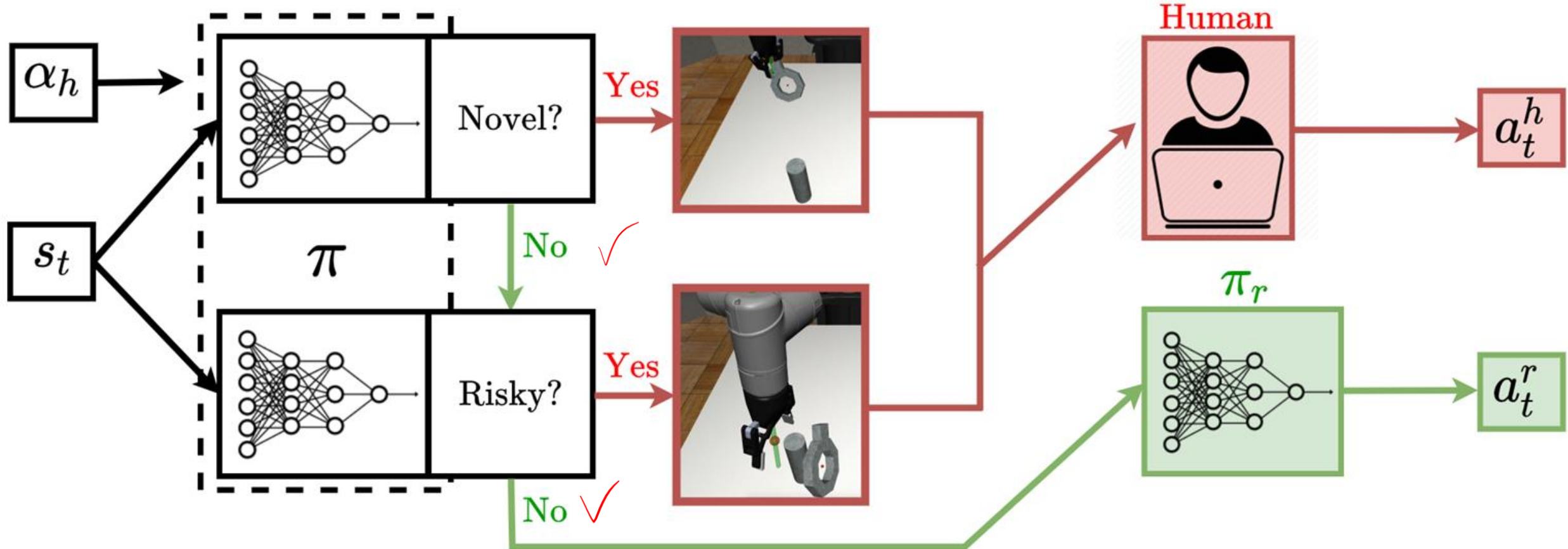
ThriftyDAgger

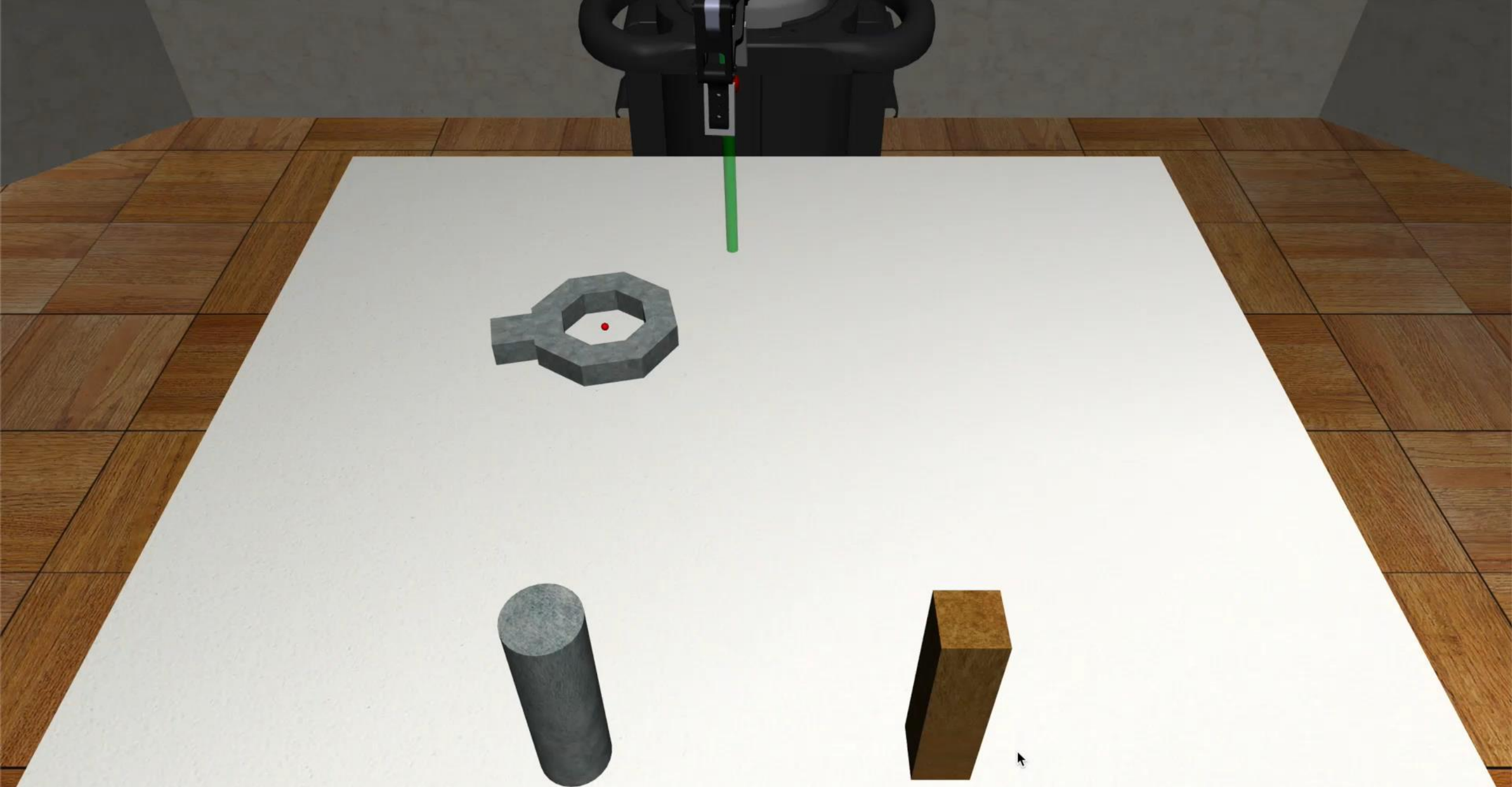


ThriftyDAgger



ThriftyDAgger



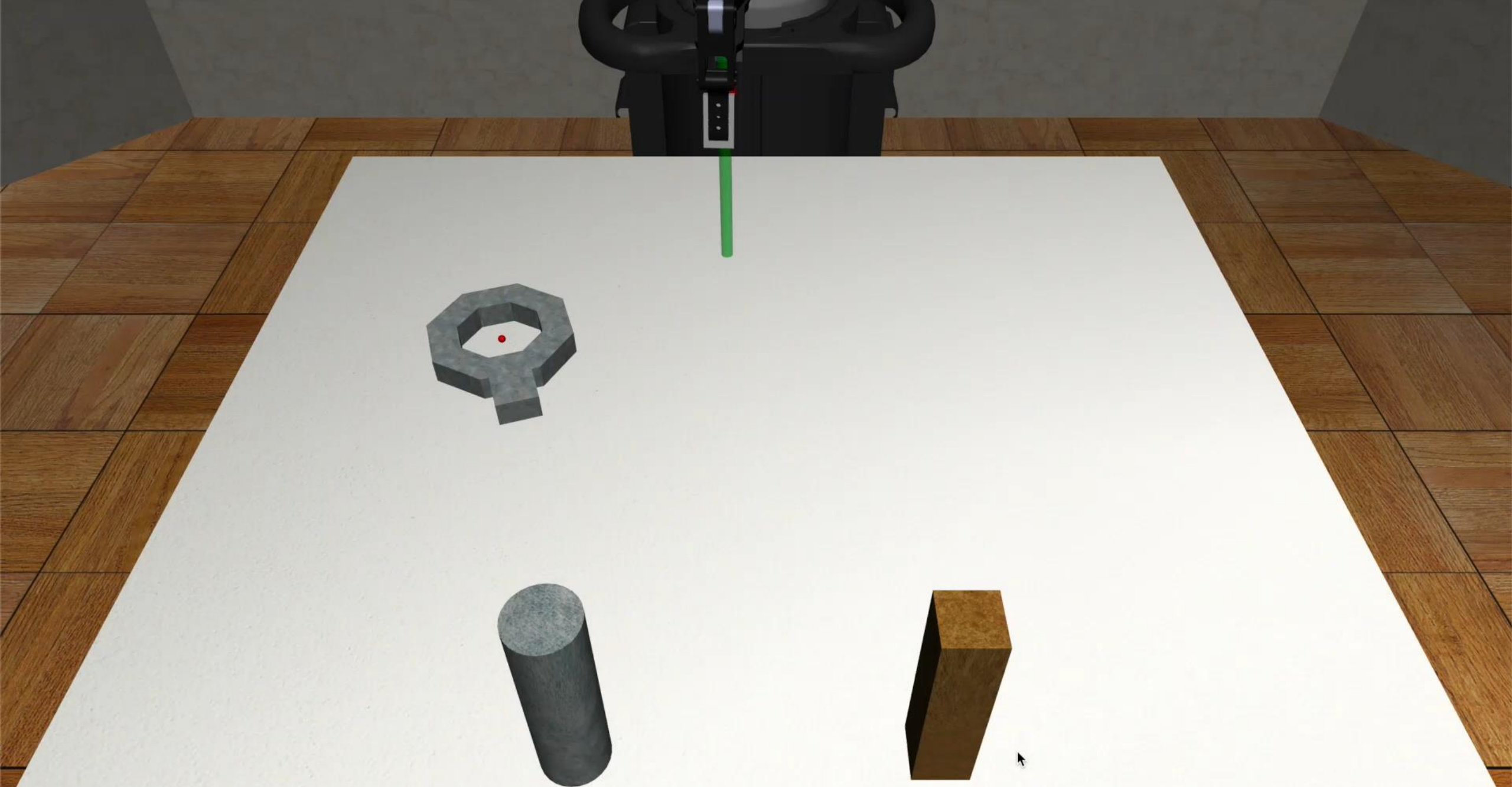


Autonomous Mode

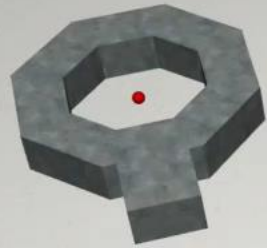


Supervisor Mode (Novel)

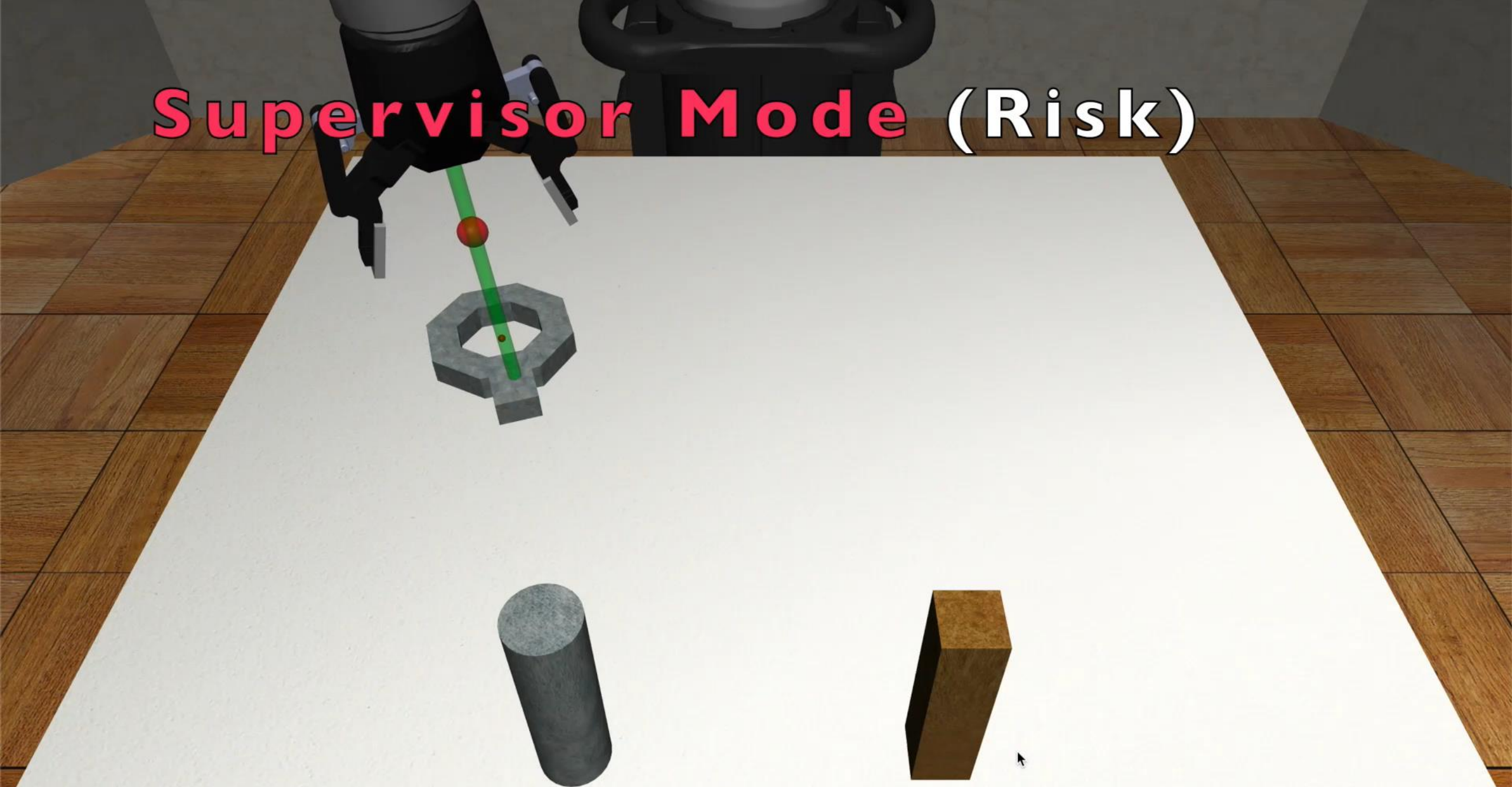




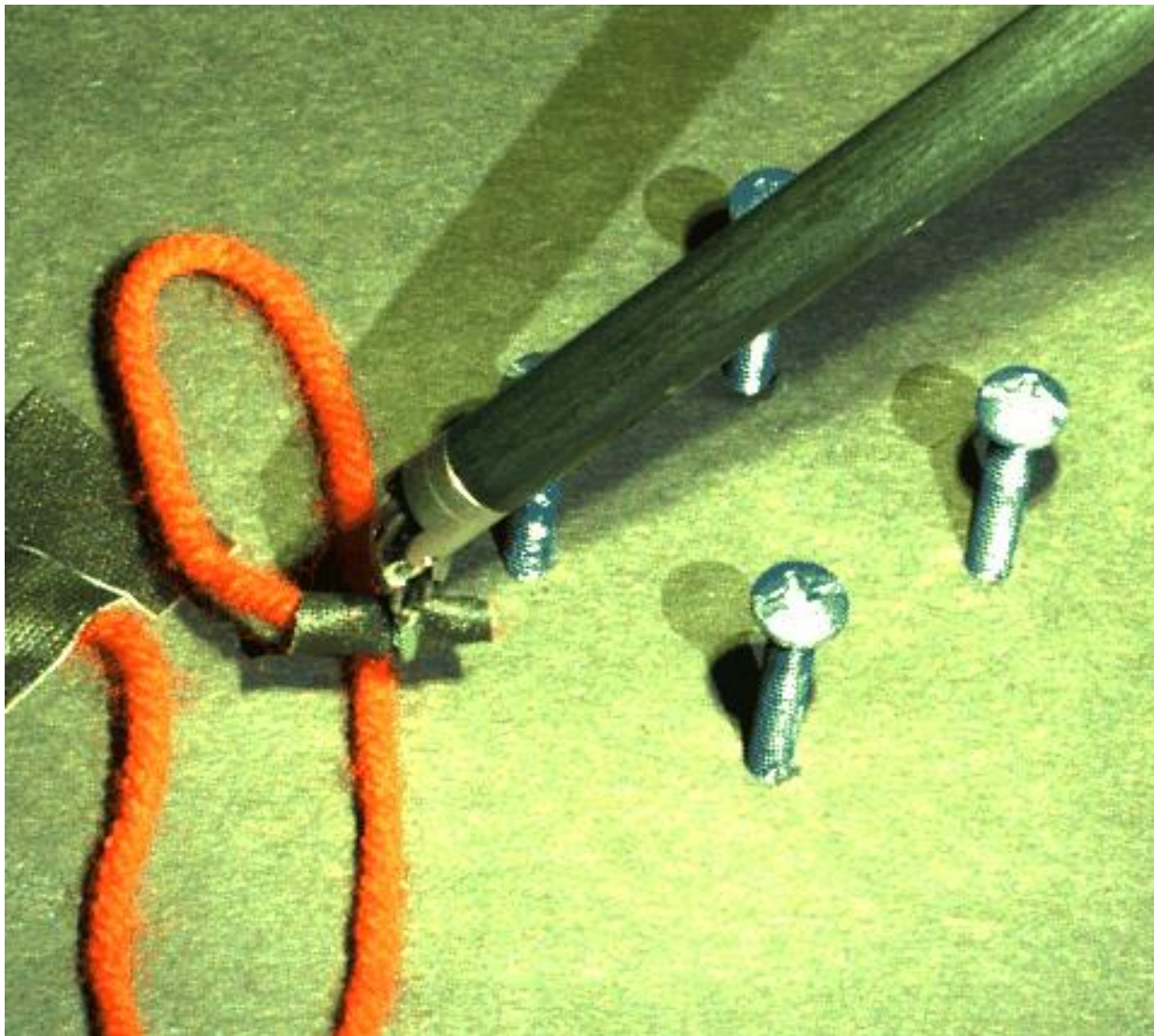
Supervisor Mode (Risk)



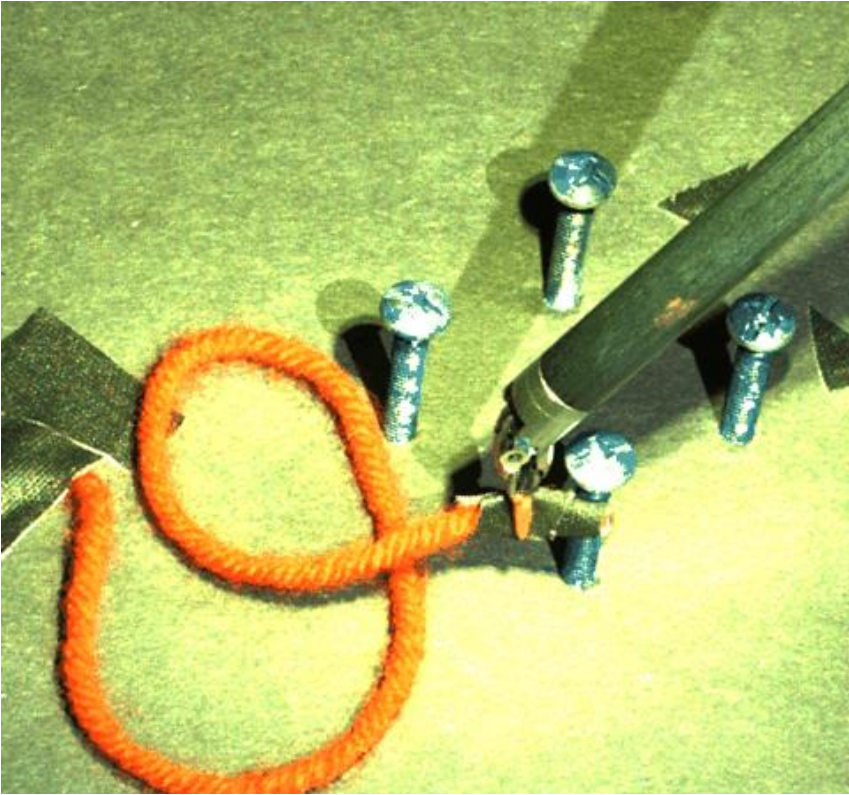
Supervisor Mode (Risk)



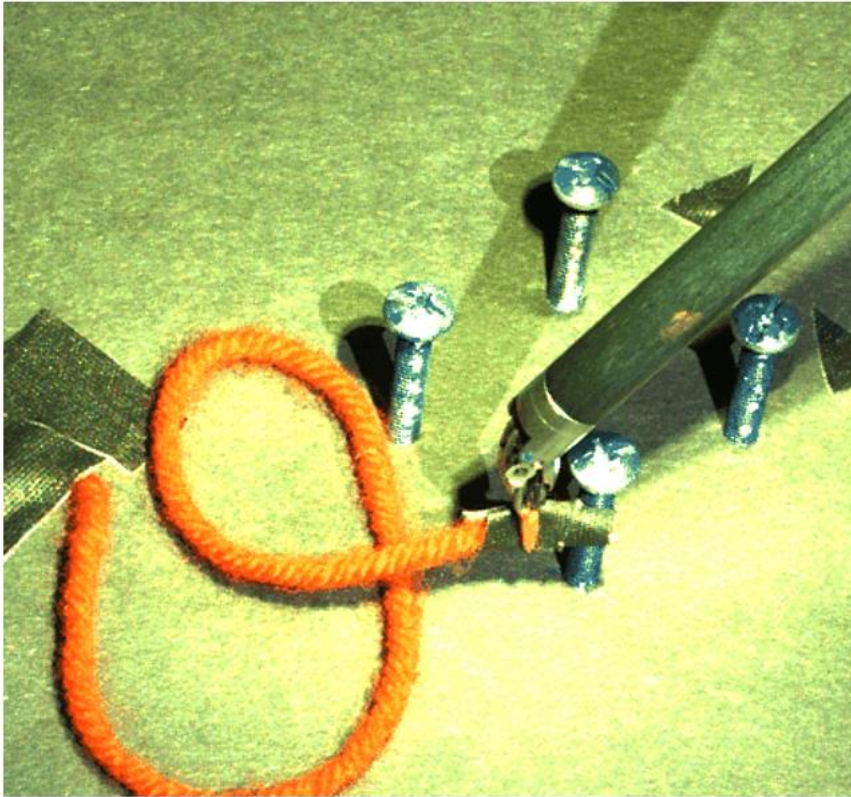
Human Demonstration



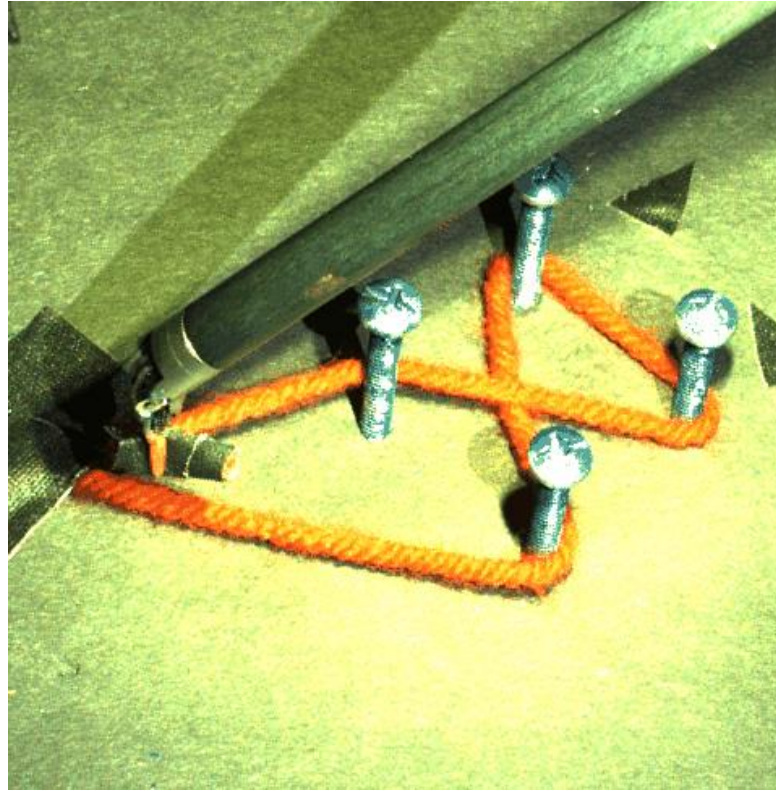
Behavior Cloning



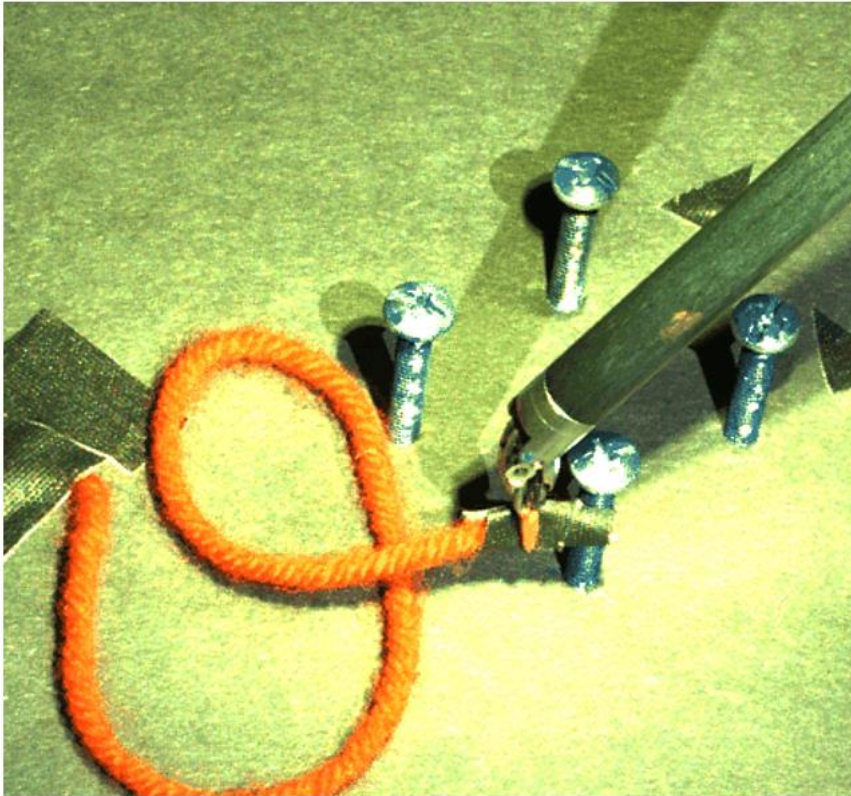
Behavior Cloning



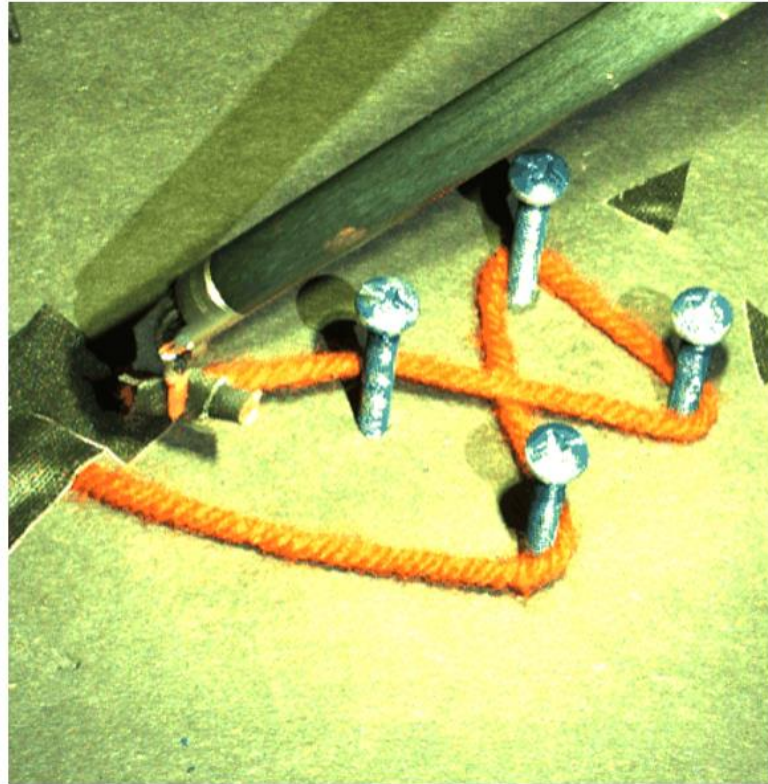
ThriftyDagger (autonomous)



Behavior Cloning



ThriftyDAgger (autonomous)



ThriftyDAgger (+human)



User Study

N=10 subjects each control 3 robots in simulation.

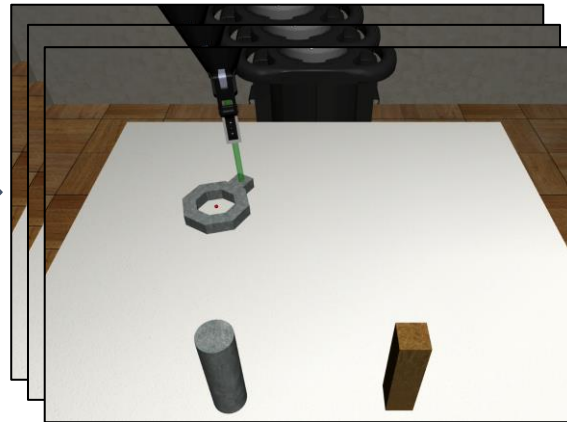
Robot-Gated

Memory: Non-Match

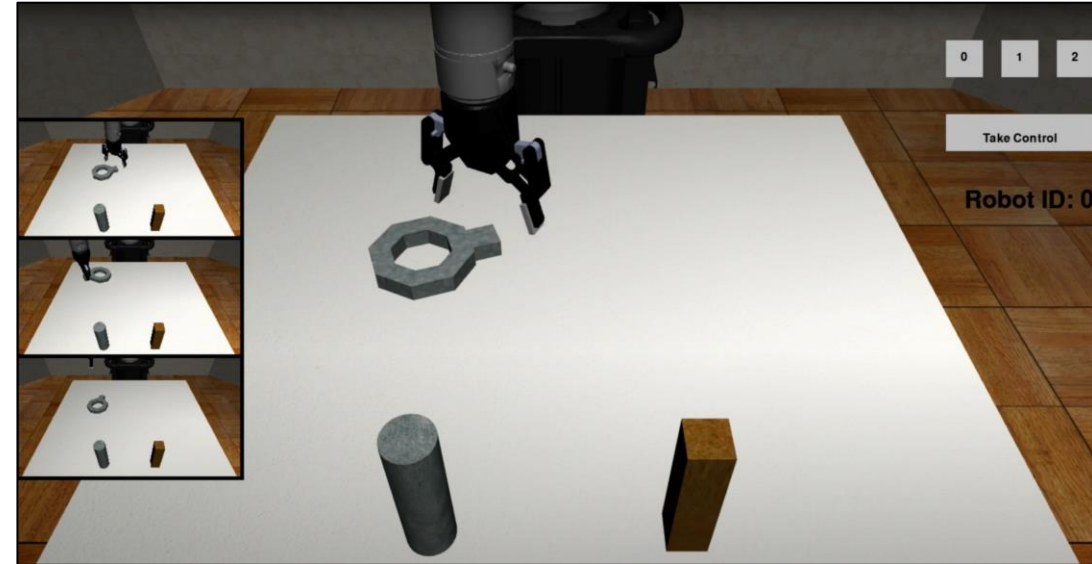
H	H	H	H	H
H	🍌	❤️	H	H
H	H	H	H	H
H	H	H	H	H

Memory: Match

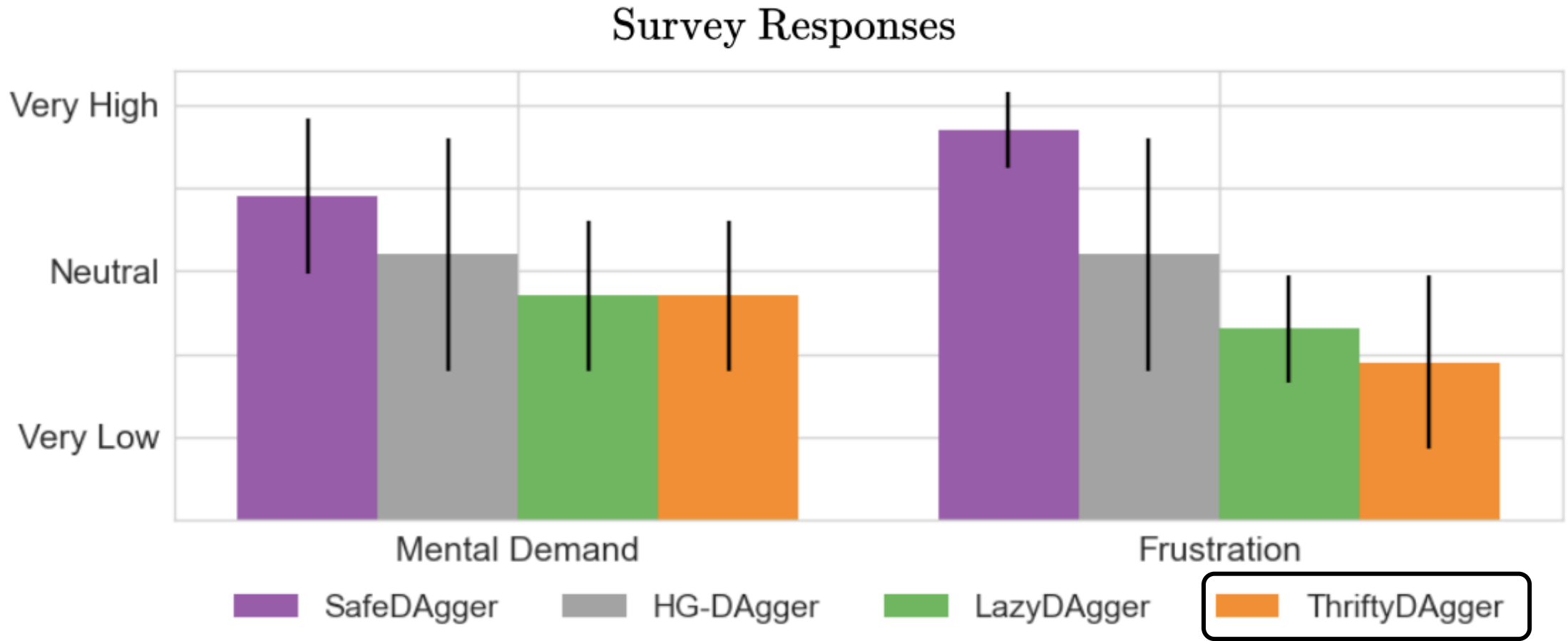
H	H	H	H	H
H	H	H	🕒	H
🕒	H	H	H	H
H	H	H	H	H



Human-Gated



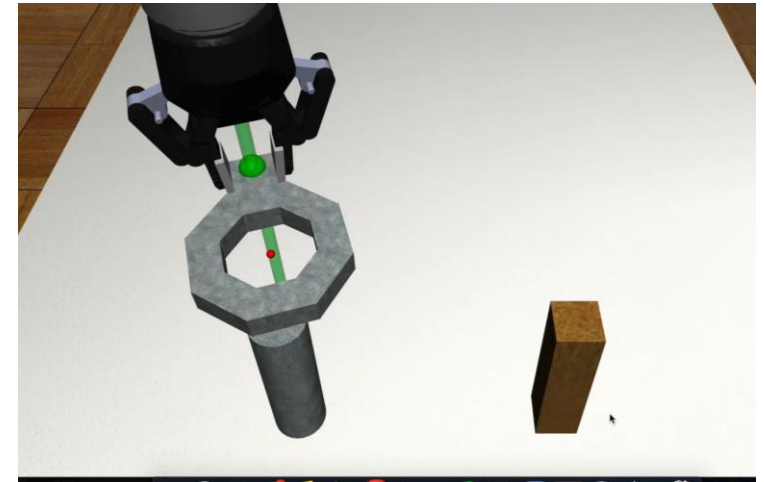
ThriftyDAgger Qualitative Results



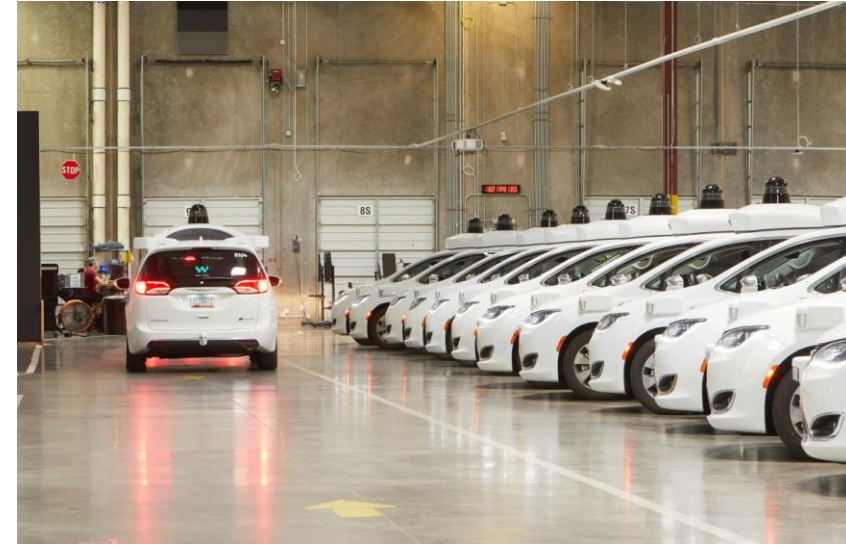
User Study Quantitative Results

ThriftyDAgger had

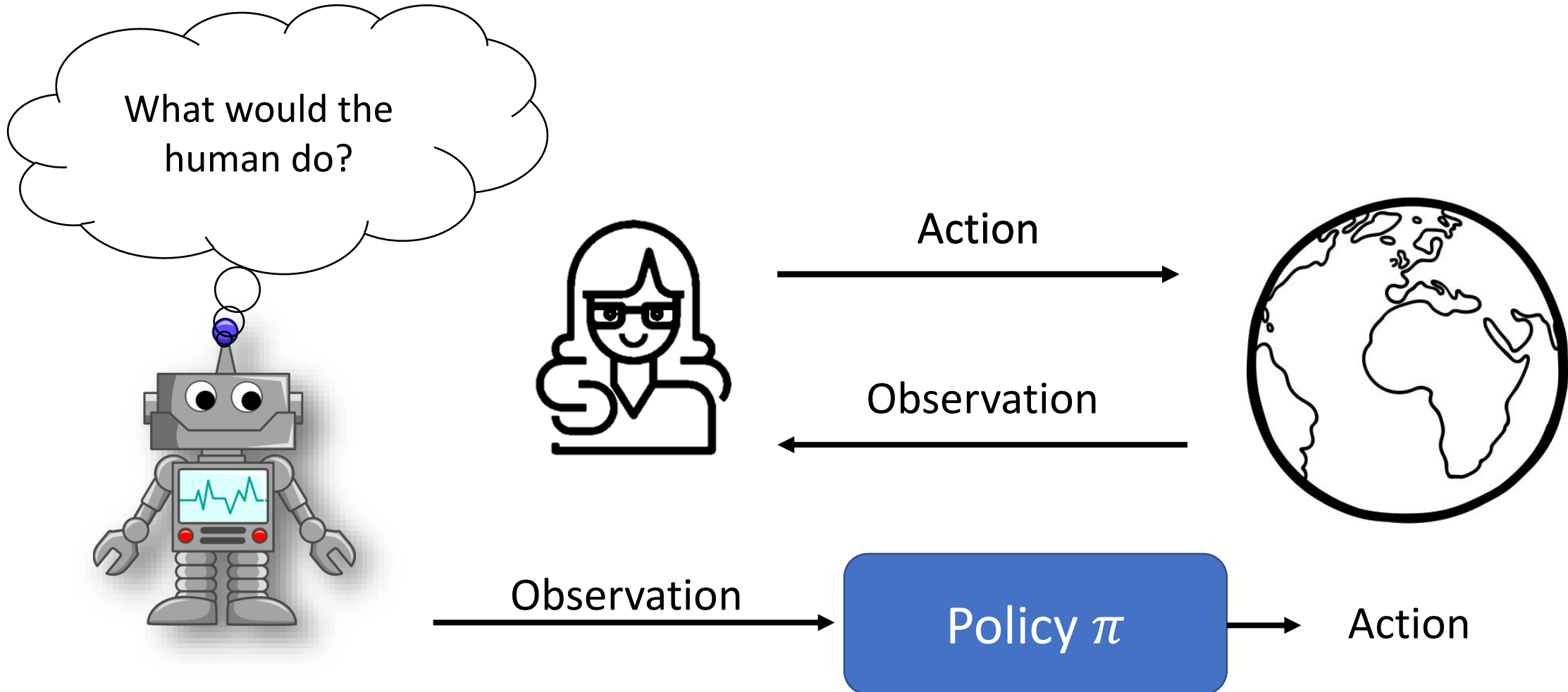
- 21% fewer human interventions
- 57% more concentration pairs found
- 80% more throughput



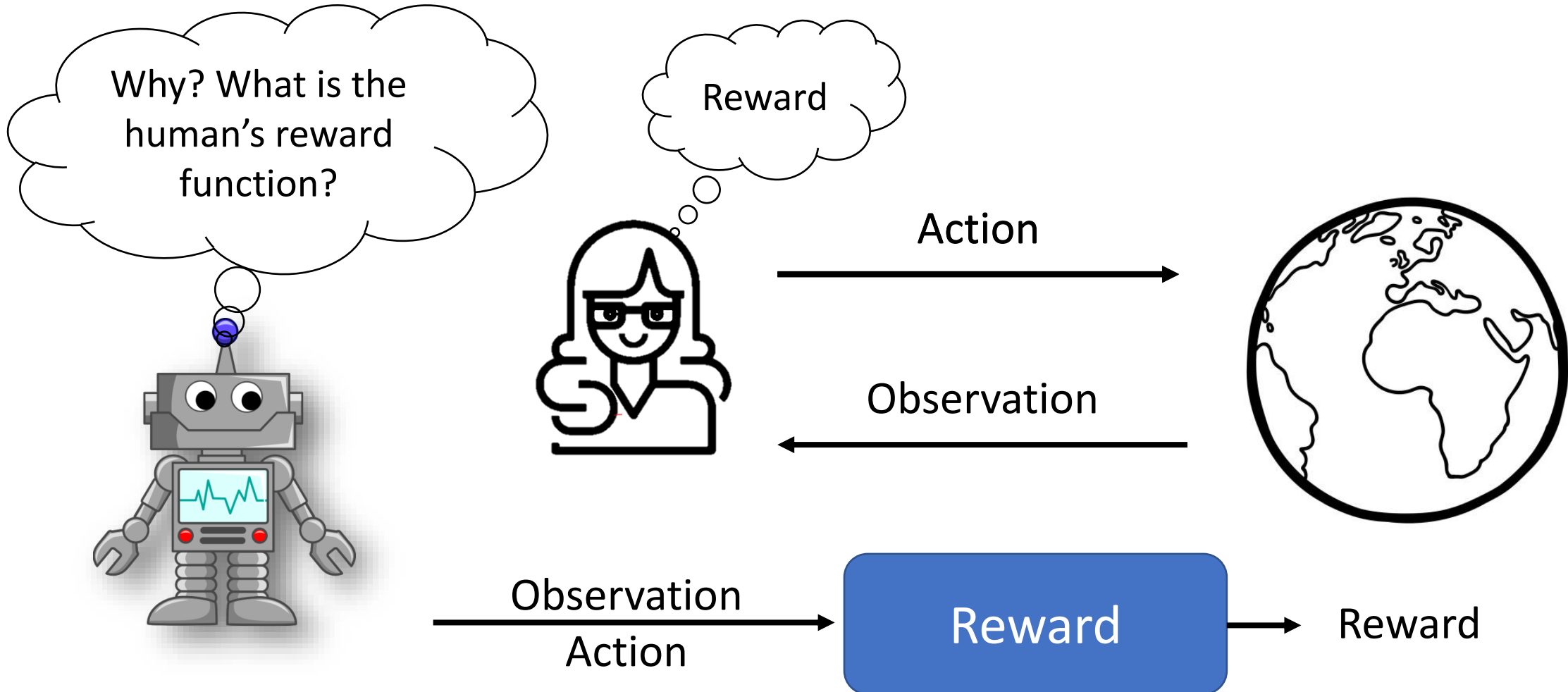
Lots of potential applications of human interaction with robot fleets!



Behavioral Cloning



Reward Learning (Inverse Reinforcement Learning)





Human Intent Inference

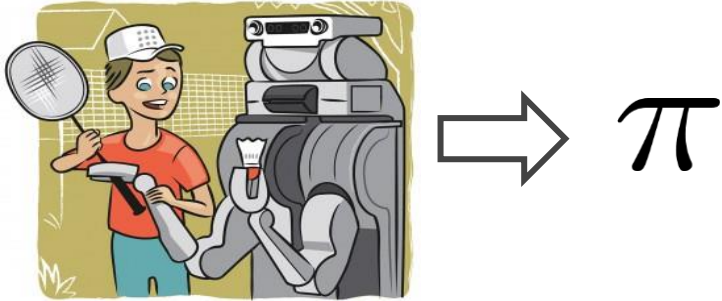


Inverse Reinforcement Learning

- Given
 - MDP without a reward function
 - Demonstrations from an optimal policy π^*
- Recover the reward function R that makes π^* optimal

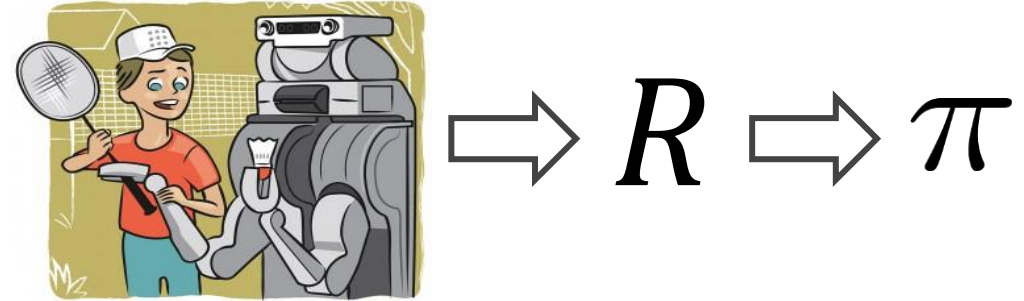
Imitation Learning

Behavioral Cloning



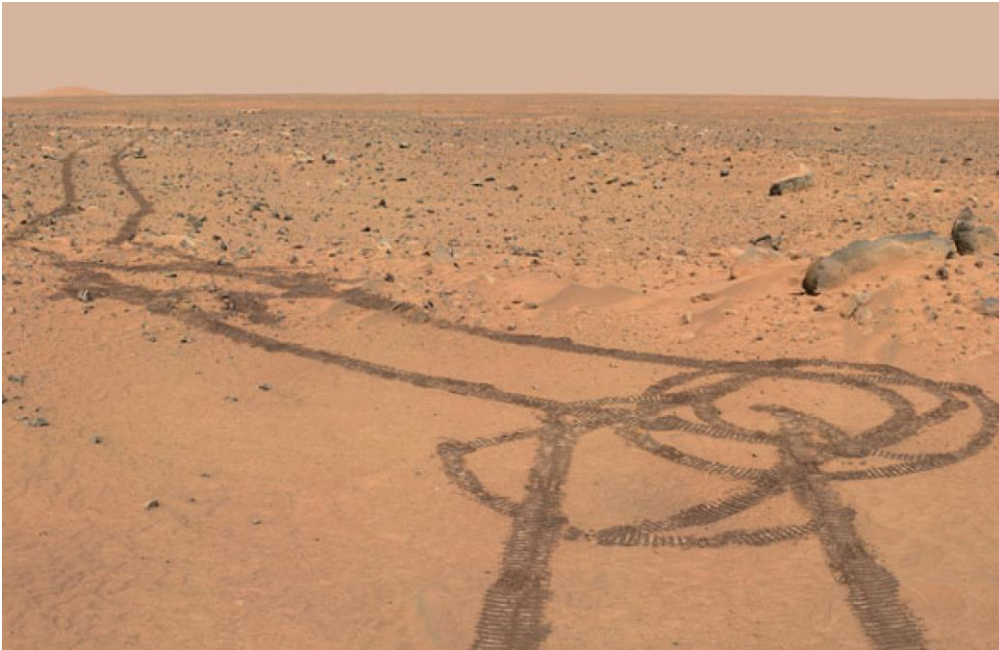
- Answers the “How?” question
- Mimic the demonstrator
- Learn mapping from states to actions
- Computationally efficient
- Compounding errors

Inverse Reinforcement Learning

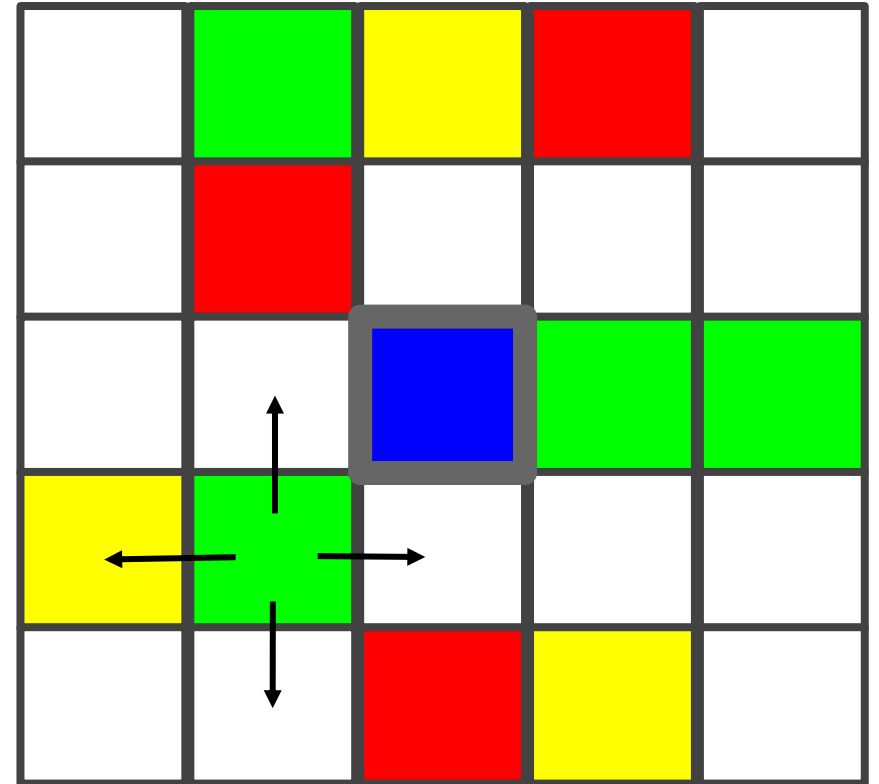


- Answers the “Why?” question
- Explain the demonstrator’s behavior
- Learn a reward function capturing the demonstrator’s intent
- Can require lots of data and compute
- Better generalization. Can recover from arbitrary states

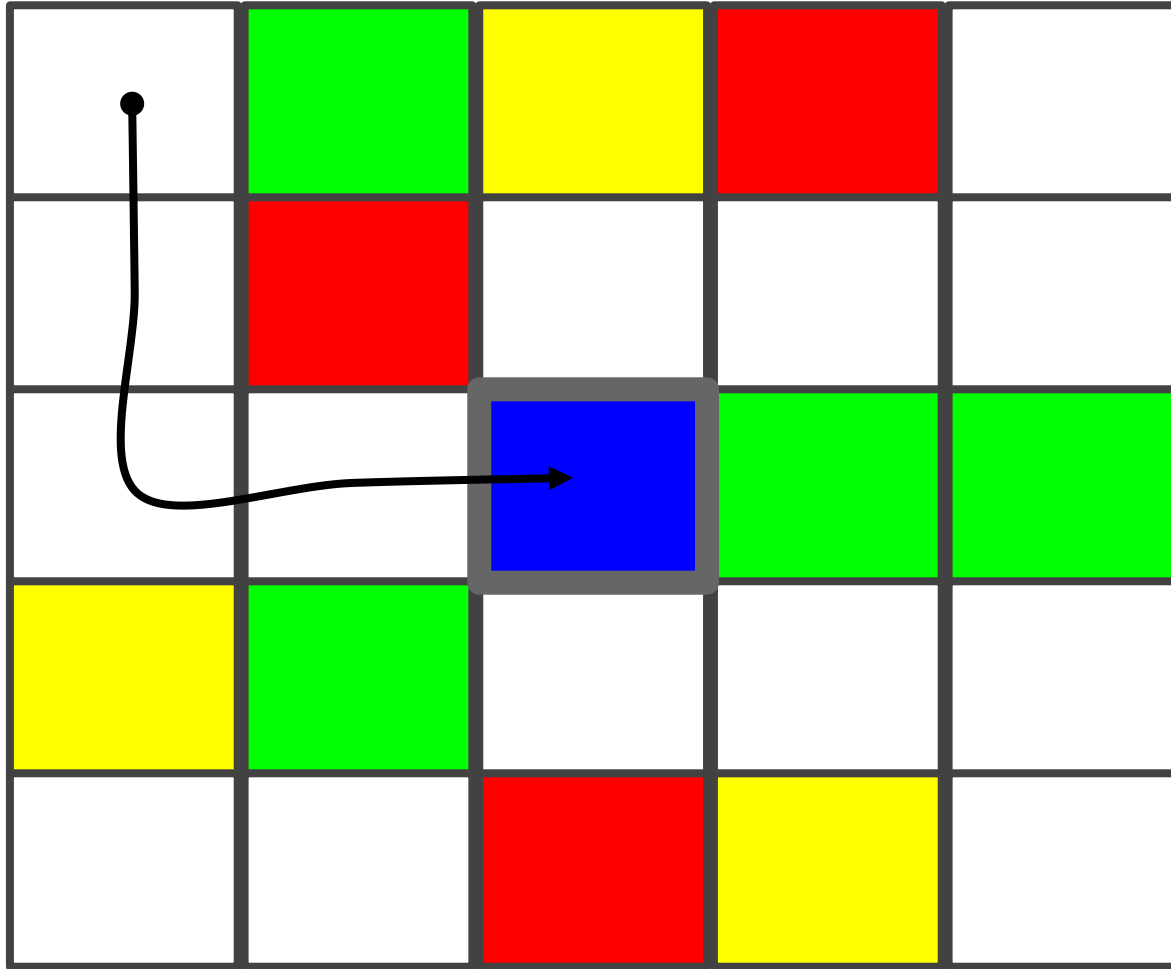
IRL Example: Teaching a robot to navigate through demonstrations



Toy version



What is the reward?



$R(\text{Blue}) = ?$

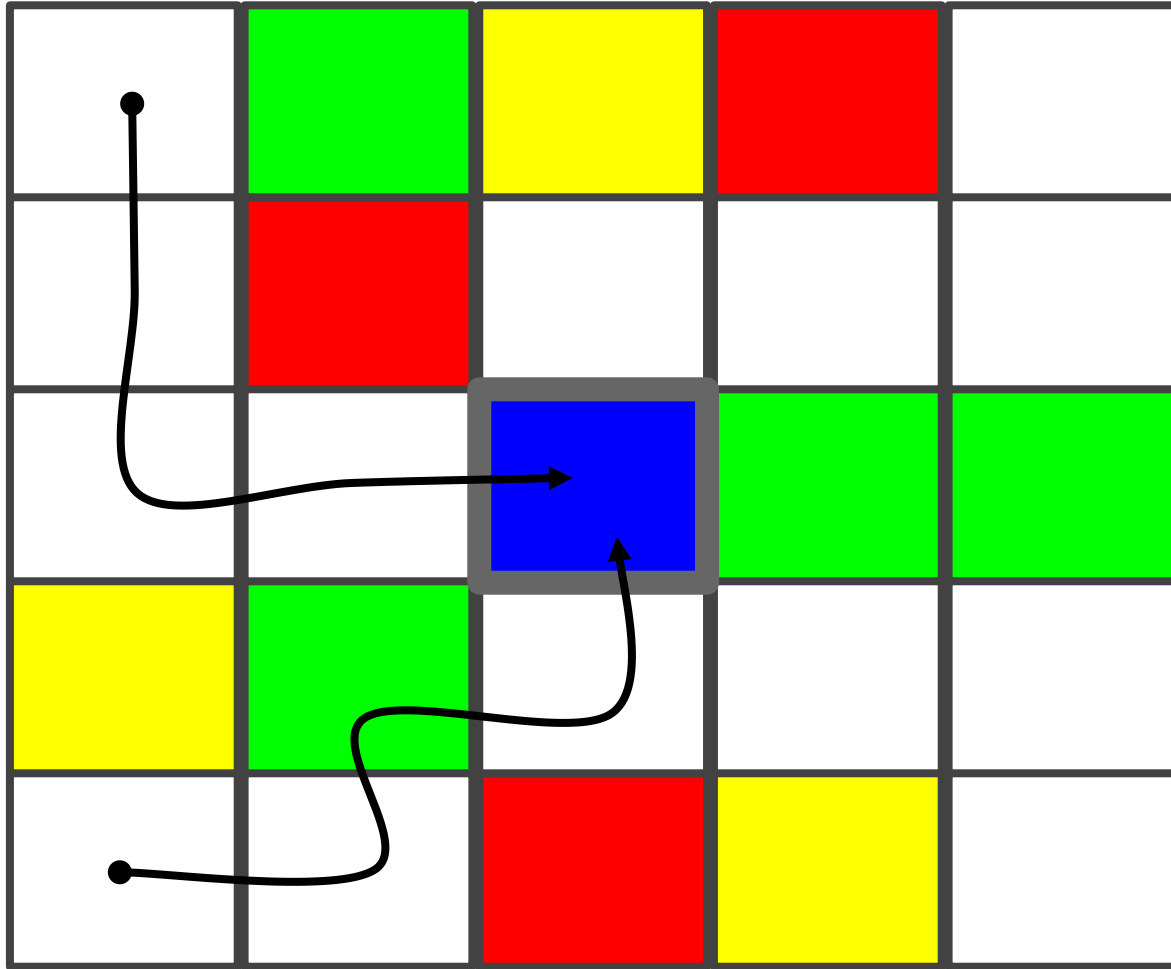
$R(\text{Yellow}) = ?$

$R(\text{Green}) = ?$

$R(\text{Red}) = ?$

$R(\text{White}) = ?$

What is the reward?



$R(\text{Blue}) = ?$

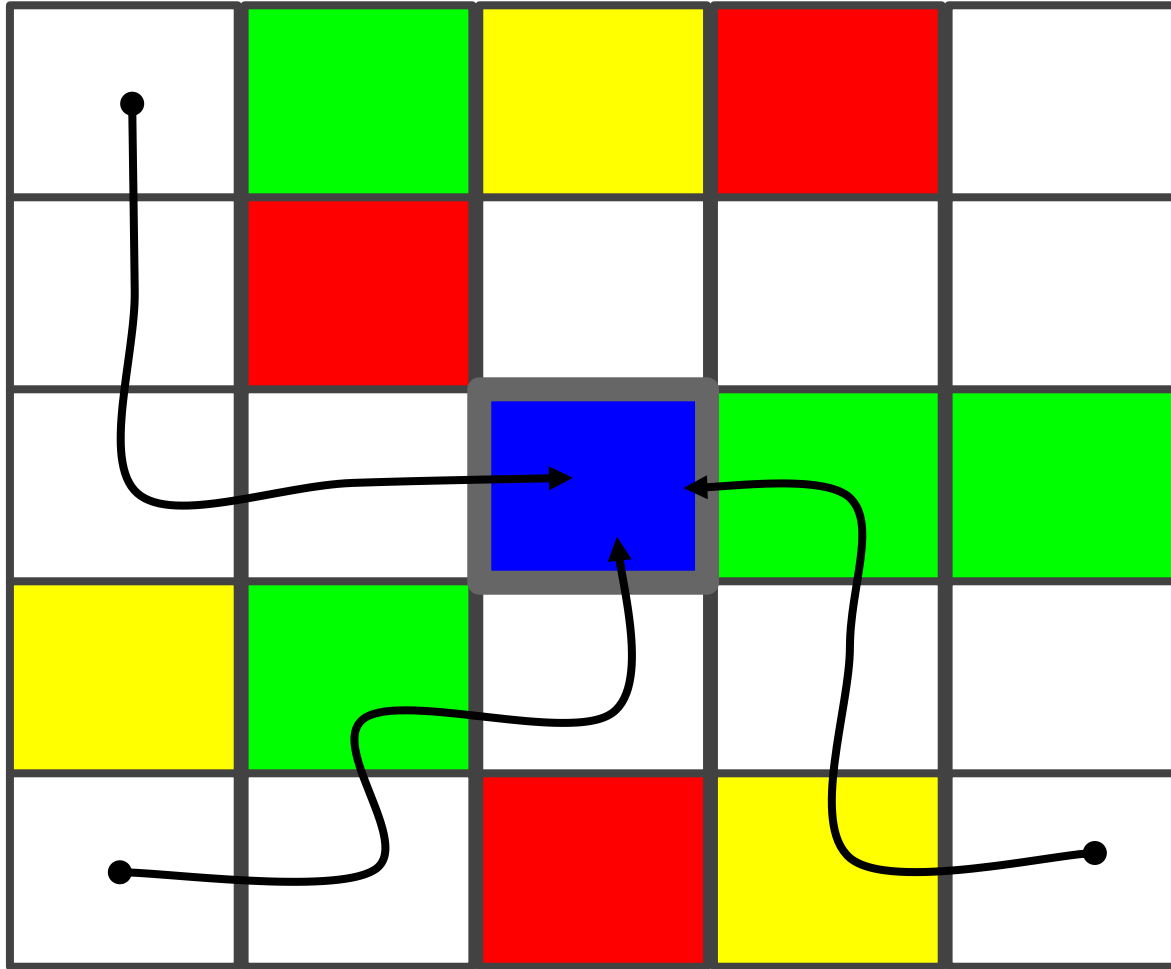
$R(\text{Yellow}) = ?$

$R(\text{Green}) = ?$

$R(\text{Red}) = ?$

$R(\text{White}) = ?$

What is the reward?



$R(\text{Blue Square}) = ?$

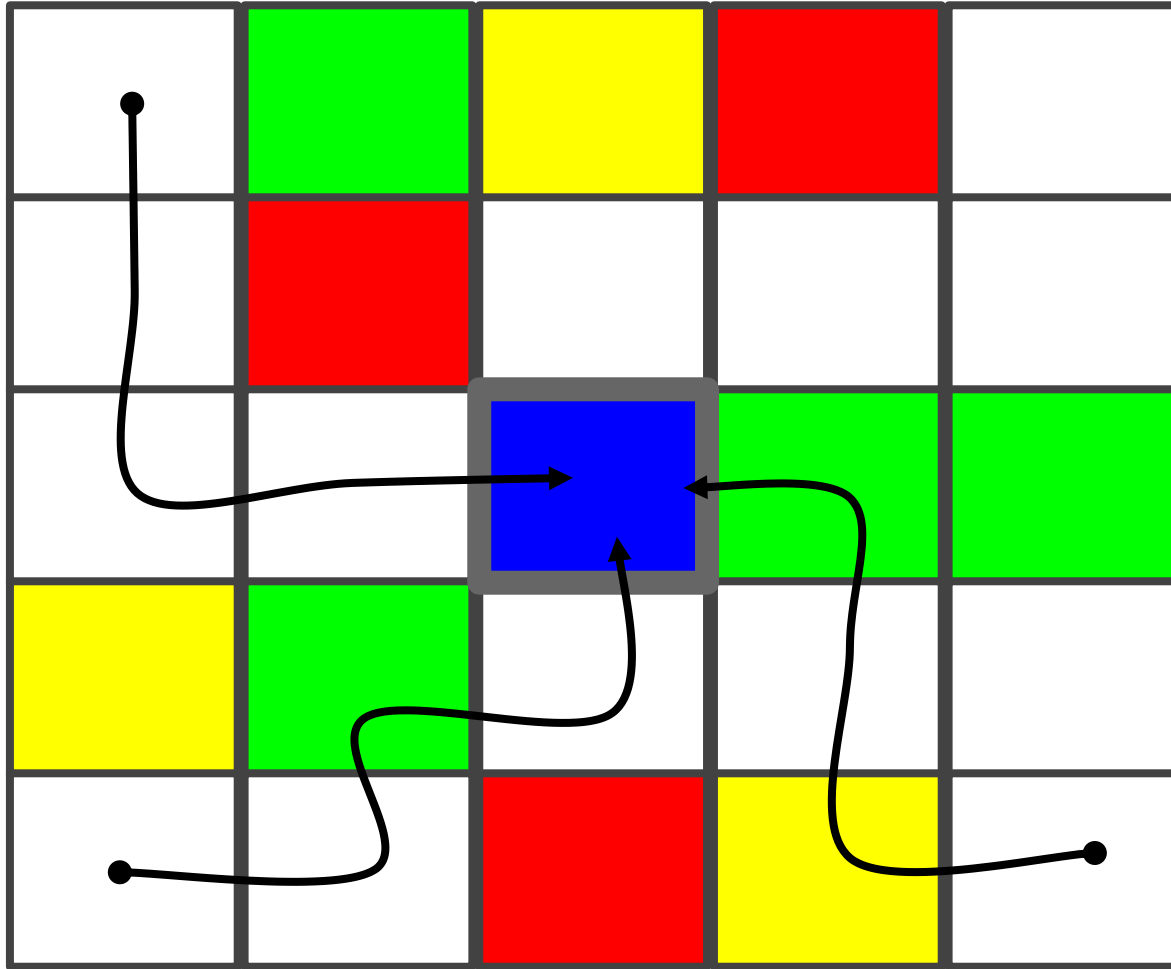
$R(\text{Yellow Square}) = ?$

$R(\text{Green Square}) = ?$

$R(\text{Red Square}) = ?$

$R(\text{White Square}) = ?$

What is the reward?



$$R(\text{Blue Square}) = +1$$

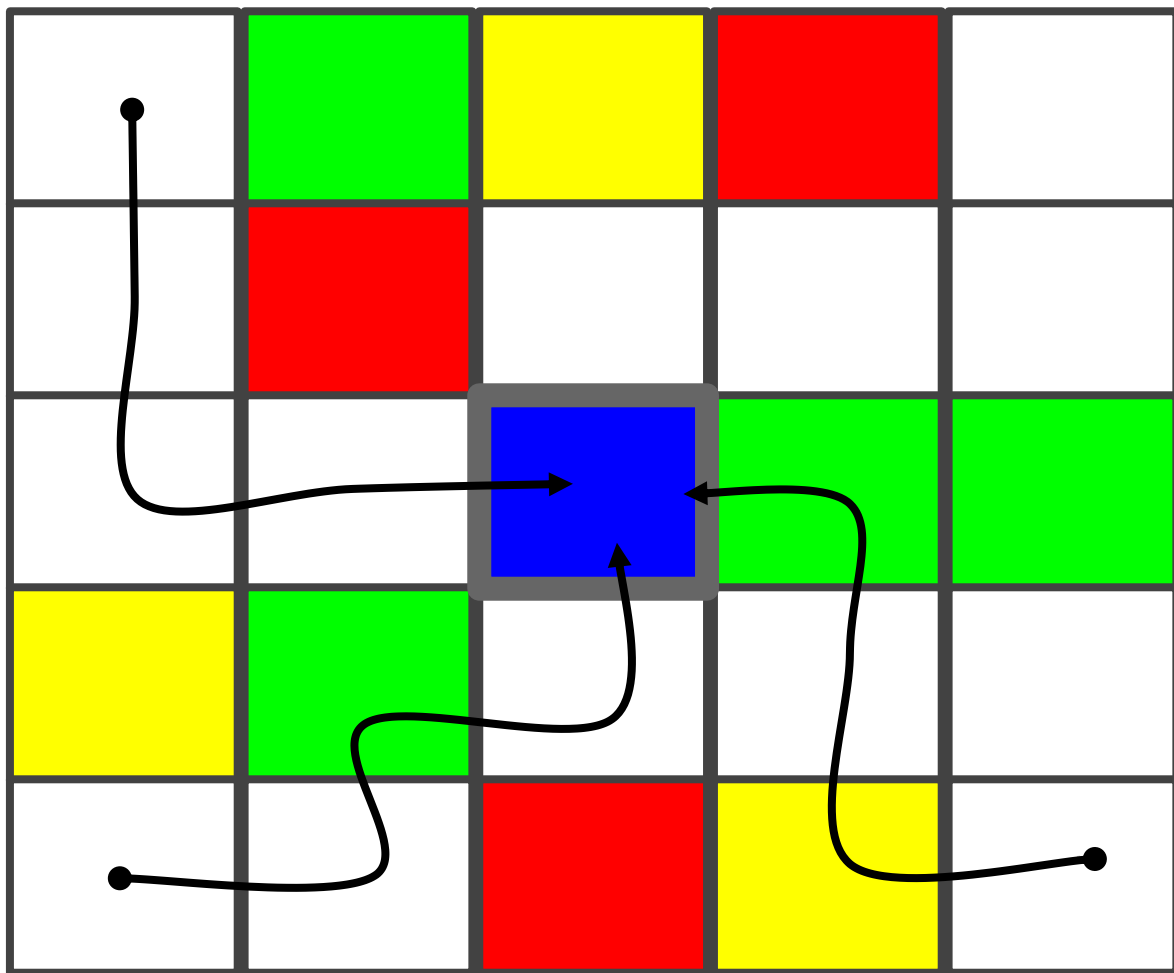
$$R(\text{Yellow Square}) = 0$$

$$R(\text{Green Square}) = 0$$

$$R(\text{Red Square}) = -1$$

$$R(\text{White Square}) = 0$$

What is the reward?



$$R(\text{Blue}) = +10$$

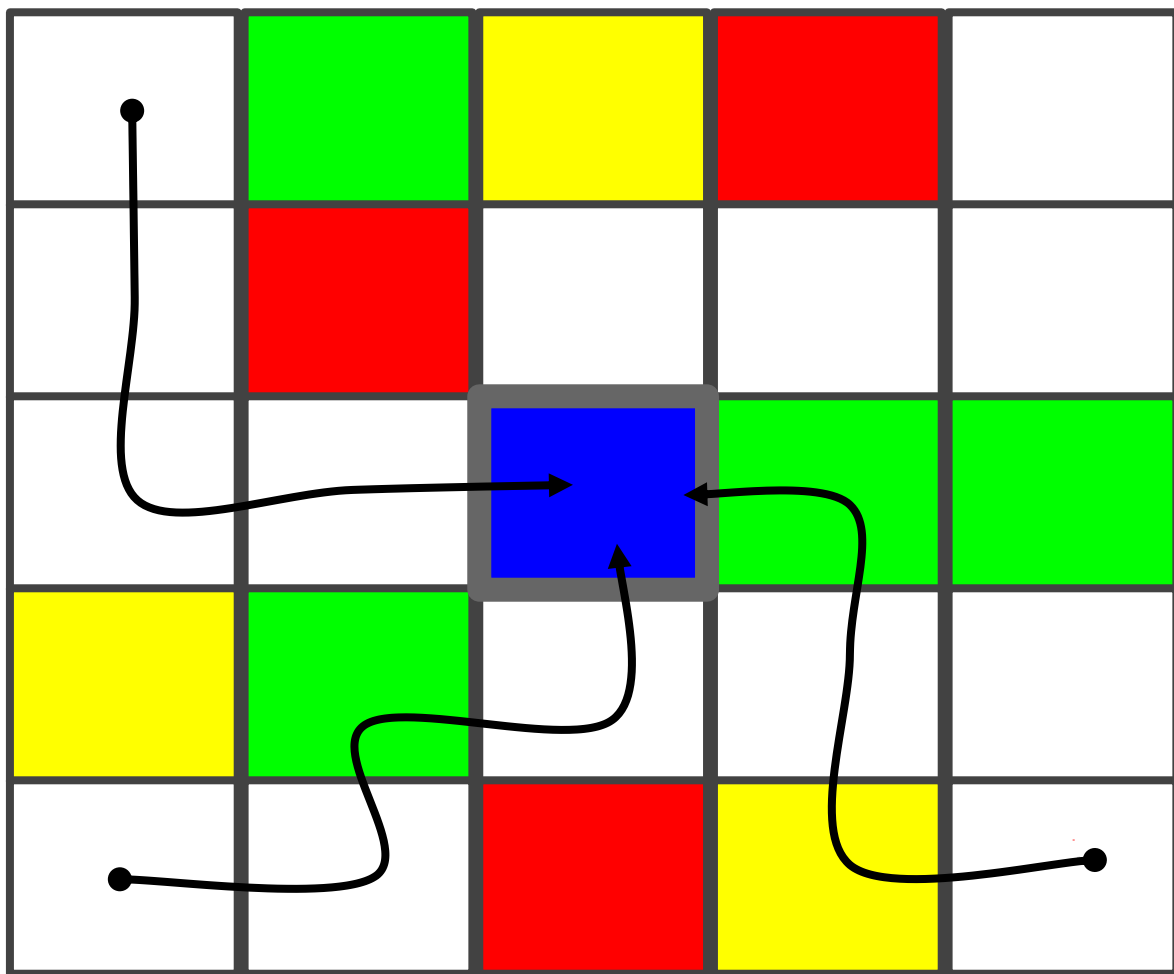
$$R(\text{Yellow}) = 0$$

$$R(\text{Green}) = 0$$

$$R(\text{Red}) = -10$$

$$R(\text{White}) = 0$$

What is the reward?



$$R(\text{Blue}) = +10$$

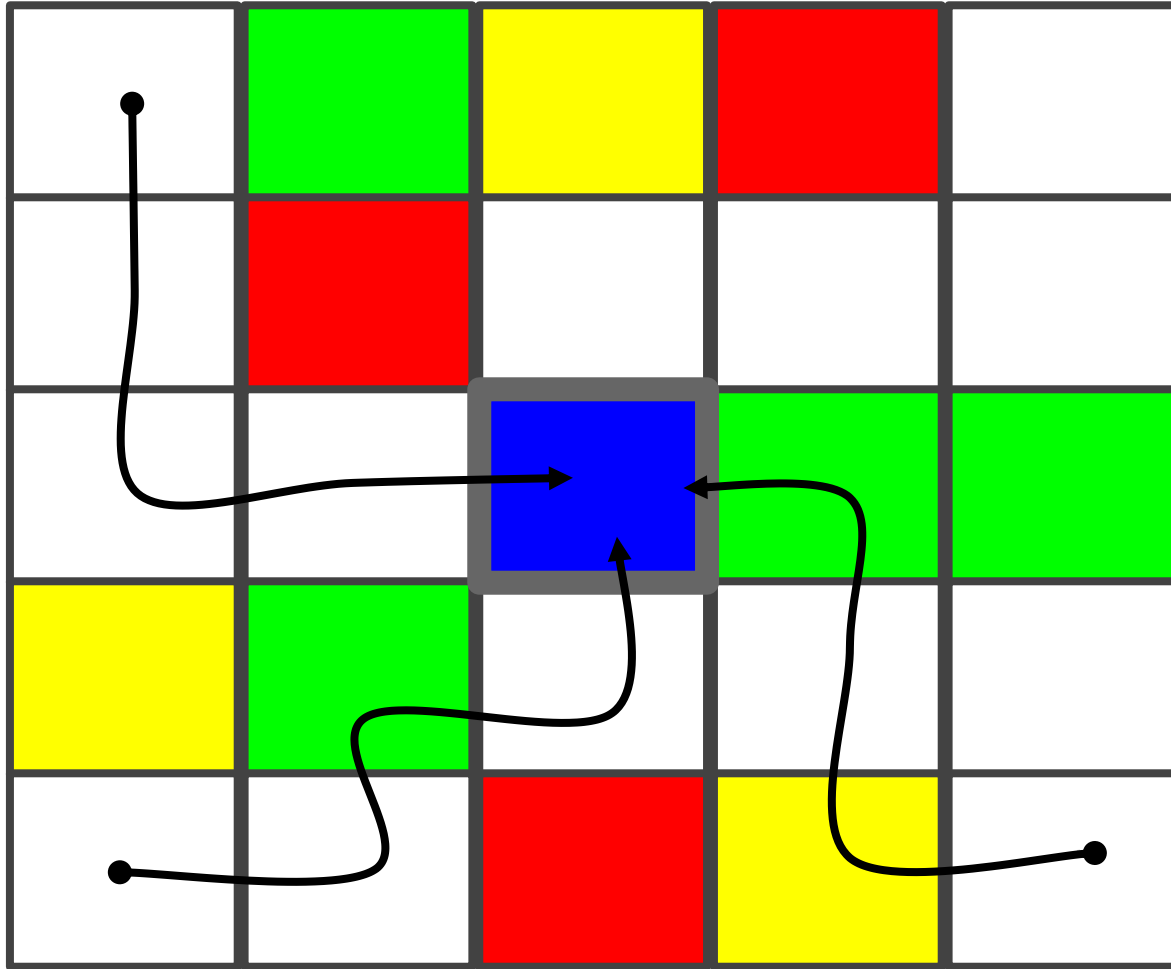
$$R(\text{Yellow}) = -1$$

$$R(\text{Green}) = -1$$

$$R(\text{Red}) = -10$$

$$R(\text{White}) = -1$$

What is the reward?



$$R(\text{blue square}) = 0$$

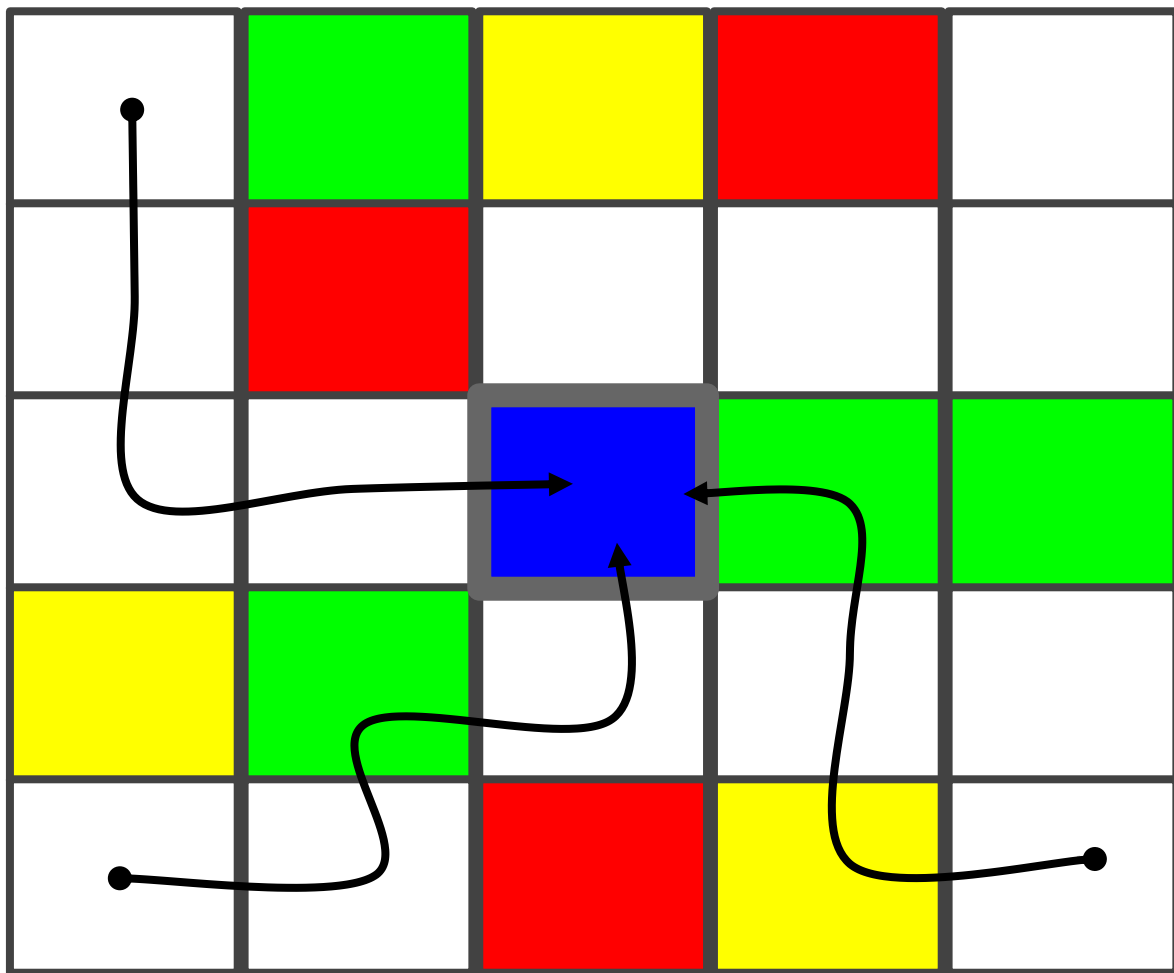
$$R(\text{yellow square}) = 0$$

$$R(\text{green square}) = 0$$

$$R(\text{red square}) = 0$$

$$R(\text{white square}) = 0$$

What is the reward?



$$R(\text{blue square}) = c$$

$$R(\text{yellow square}) = c$$

$$R(\text{green square}) = c$$

$$R(\text{red square}) = c$$

$$R(\text{white square}) = c$$

Inverse Reinforcement Learning Formalism

- Given
 - MDP without a reward function
 - Demonstrations from an optimal policy π^*
- Recover a reward function R that makes π^* optimal
- Ill-Posed Problem
 - Infinite number of reward functions that can make π^* optimal
 - Trivial all zero reward
 - Constant reward
 - $aR + c$ (positive scaling $a > 0$, and affine shifts)

Basic IRL Algorithm

- Start with demonstrations, D
- Guess initial reward function R_0
- $\hat{R} = R_0$
- Loop:
 - Solve for optimal policy $\pi_{\hat{R}}^*$
 - Compare D and $\pi_{\hat{R}}^*$
 - Update $\hat{R}$ to try and make D and $\pi_{\hat{R}}^*$ more similar

Flashback: Approximate Q-Learning

$$Q(s, a) = w_1 f_1(s, a) + w_2 f_2(s, a) + \dots + w_n f_n(s, a)$$

- Q-learning with linear Q-functions:

$$\text{transition} = (s, a, r, s')$$

$$\text{difference} = \left[r + \gamma \max_{a'} Q(s', a') \right] - Q(s, a)$$

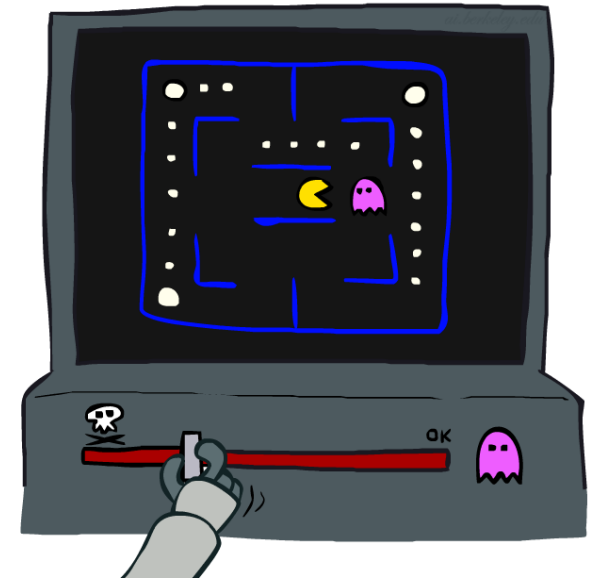
$$Q(s, a) \leftarrow Q(s, a) + \alpha [\text{difference}]$$

$$w_i \leftarrow w_i + \alpha [\text{difference}] f_i(s, a)$$

Exact Q's

Approximate Q's

- Intuitive interpretation:
 - Adjust weights of active features
 - E.g., if something unexpectedly bad happens, blame the features that were on: disprefer all states with that state's features
- Formal justification: online least squares



Feature count matching

- Assume the reward function is a linear combination of features:

$$R(s) = \mathbf{w}^T \phi(s)$$

- Value function becomes linear combination of (discounted) feature expectations:

$$V_R^\pi = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R(s_t) \right]$$

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$$V_R^\pi = \mathbf{w}^T \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t \phi(s_t) \right] = \mathbf{w}^T \mu_\pi$$

Inverse reinforcement learning: feature matching

(Abbeel and Ng 2004, Syed and Schapire 2007)

- If $\|\mathbf{w}\|_1 \leq 1$, then

$$|x^\top y| \leq \|x\|_1 \|y\|_\infty$$

$$\begin{aligned} V_R^{\pi^*} - V_R^{\pi_{\text{robot}}} &= \mathbf{w}^T (\mu_{\pi^*} - \mu_{\pi_{\text{robot}}}) \\ &\leq \|\mu_{\pi^*} - \mu_{\pi_{\text{robot}}}\|_\infty \end{aligned}$$

- If feature expectations match, then expected returns are identical.
- Idea: Can we update the reward guess $\hat{R}$ so the feature counts get closer?

Problem: Many different policies can lead to same expected feature counts

Maximum Entropy IRL (Ziebart et al. 2008)

- Collect M demonstrations $D = \{\tau_1, \dots, \tau_M\}$
- Initialize reward weights $\mathbf{w}$
- **Loop**
 - Solve for (soft) optimal policy $\pi(a|s)$ via Value Iteration
 - Solve for expected feature counts of $\pi(a|s)$
 - Compute weight update

$$\mathbf{w} \leftarrow \mathbf{w} + \alpha(\mu_D - \mu_\pi)$$

$$P(\tau) = \frac{e^{R_{\mathbf{w}}(\tau)}}{Z}$$

$$R(s) = \mathbf{w}^T \phi(s)$$

Watch This: Scalable Cost-Function Learning for Path Planning in Urban Environments

Markus Wulfmeier¹, Dominic Zeng Wang¹ and Ingmar Posner¹

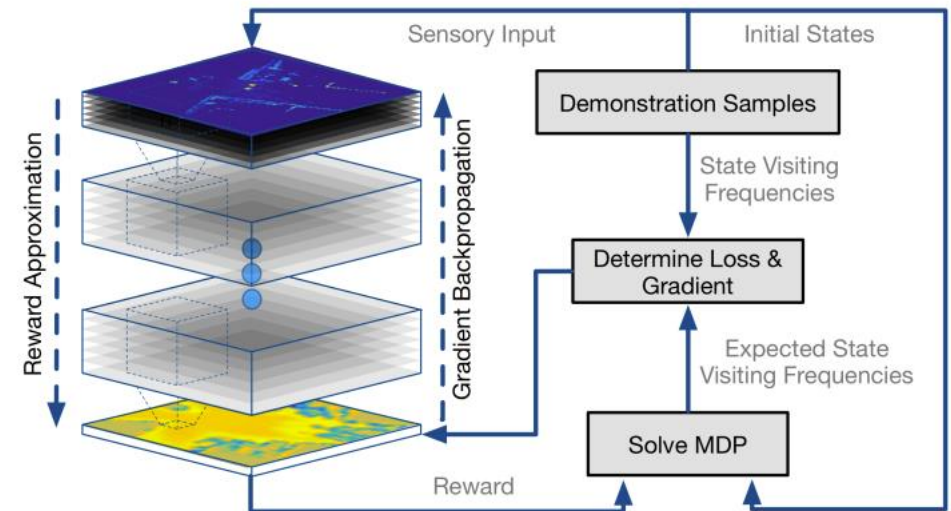
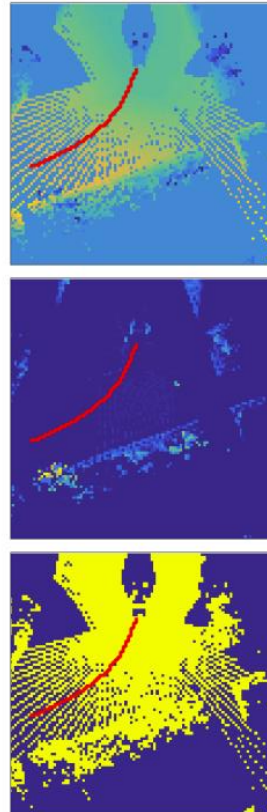


Fig. 1: Schema for training neural networks in the Maximum Entropy paradigm for IRL.

Another way to look at MaxEnt IRL

$$P(\tau) = \frac{e^{R_w(\tau)}}{Z} \quad Z = \int e^{R_w(\tau)} d\tau$$

- Maximum Likelihood Estimation
- Find reward function that maximizes the log likelihood of the demonstration trajectories:

$$\max_{\theta} \frac{1}{N} \sum_{\tau \in D} R_w(\tau) - \log Z$$

How to avoid fully solving MDP

$$\max_{\theta} \frac{1}{N} \sum_{\tau \in D} R_w(\tau) - \log Z \qquad Z = \int e^{R_w(\tau)} d\tau$$

- Estimate Z with a finite set of trajectories Z_{τ} .
- Loop:
 - Update parameters w so demonstrations have higher reward than trajectories in Z_{τ} .
 - Update Z_{τ}

Finn et al. "Guided Cost Learning." 2016



What if we don't want just a single reward estimate?

- Can we get a samples from the full Bayesian posterior?

$$P(R|D) \propto P(D|R)P(R)$$

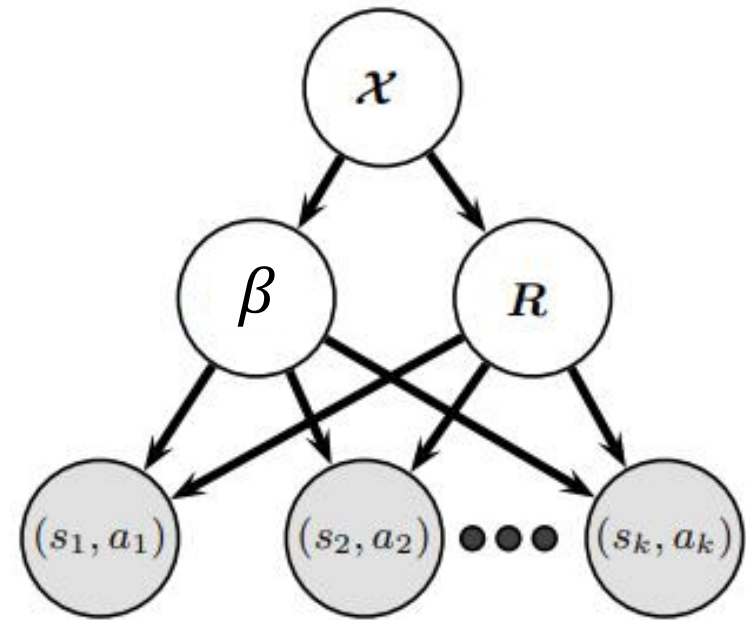


Figure 2: The BIRL model

Bayesian Inverse Reinforcement Learning

(Ramachandran and Amir 2007)

- Assume demonstrator is Boltzmann rational
 - Demonstrator follows a softmax policy with inverse temperature c

$$P(D|R) = \prod_{(s,a) \in D} \frac{e^{\beta Q^*(s,a,R)}}{\sum_{b \in A} e^{\beta Q^*(s,b,R)}}$$

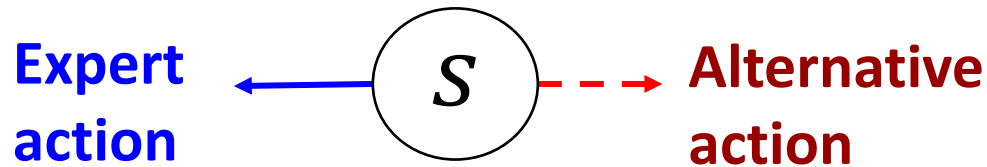
$Q^*(s, a, R) =$ How much reward will I expect to see if I take action a in state s and act optimally thereafter.

Bayesian Inverse Reinforcement Learning

(Ramachandran and Amir 2007)

- Assume demonstrator is Boltzmann rational
 - Demonstrator follows a softmax policy with inverse temperature β

$$P(D|R) = \prod_{(s,a) \in D} \frac{e^{\beta Q^*(s,a,R)}}{\sum_{b \in A} e^{\beta Q^*(s,b,R)}}$$



$$P((s, \leftarrow) | R) = \frac{e^{Q^*(s, \leftarrow, R)}}{e^{Q^*(s, \leftarrow, R)} + e^{Q^*(s, \dashrightarrow, R)}}$$

Bayesian Inverse Reinforcement Learning

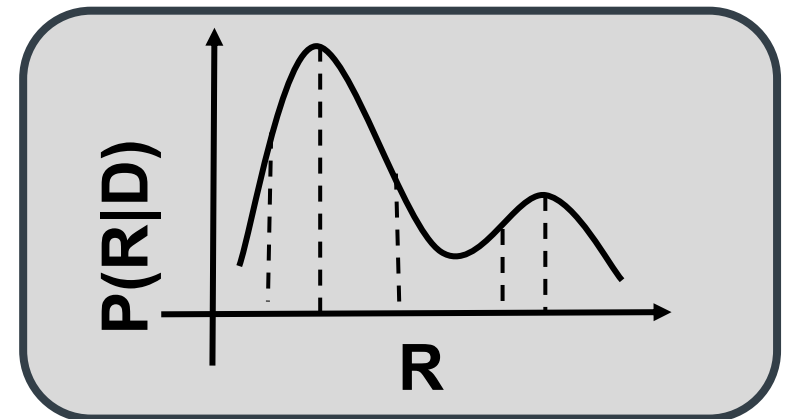
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$$P(D|R) = \prod_{(s,a) \in D} \frac{e^{\beta Q^*(s,a,R)}}{\sum_{b \in A} e^{\beta Q^*(s,b,R)}}$$

- Perform Bayesian inference (MCMC) to sample from posterior distribution

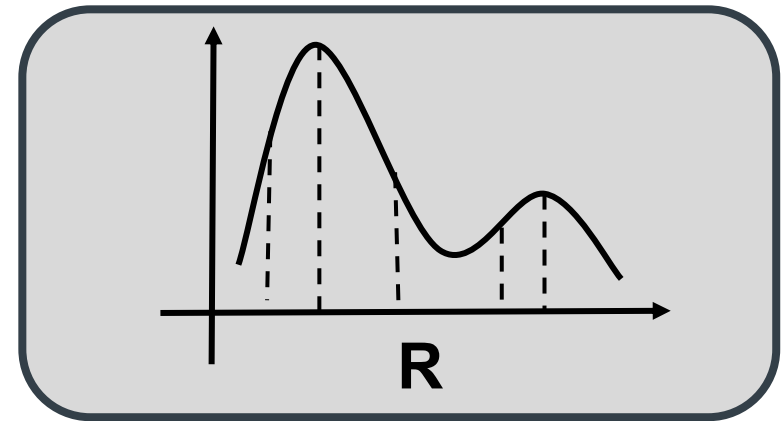
$$P(R|D) \propto P(D|R)P(R)$$



Applications of Bayesian IRL

- Active Learning
- Uncertainty Estimation
- Demonstration Sufficiency

$$P(R|D) \propto P(D|R)P(R)$$





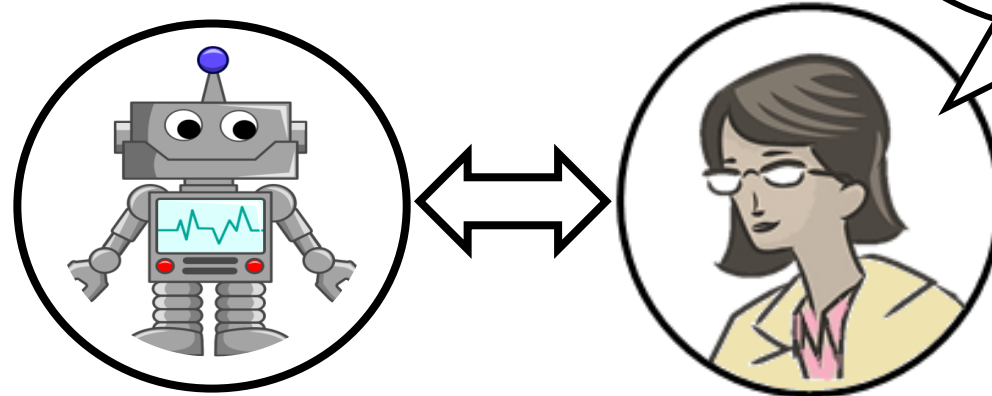
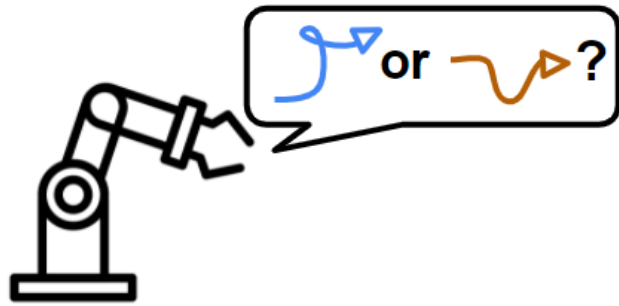
Autonomous Assessment of Demonstration Sufficiency via Bayesian Inverse Reinforcement Learning

Tu (Alina) Trinh
University of
California, Berkeley

Haoyu Chen
University of Utah

Daniel S. Brown
University of Utah

How do we learn from diverse types of feedback?



What if I want to learn rewards from more than demonstrations?

Reward-rational (implicit) choice: A unifying formalism for reward learning

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²University of California, Berkeley

Next time: Learning from pairwise preferences!